Analysis of the ‘Health-Emotion Gap’: Biological Longevity vs. Emotional Well-being

Abstract

This project explores the divergence between objective health indicators and subjective perceptions of happiness. Our main research question is whether an increase in healthy life expectancy worldwide reflects a higher degree of happiness for the global population. Our research demonstrates that, by and large, this is not the case. Indeed, in the vast majority of the countries included in the World Happiness Report 2024 (our primary dataset), the population is subject to what we define as the ‘Health-Emotion Gap.’ This metric measures the degree of unhappiness in a country relative to its healthy life expectancy, with values ranging between -1.00 and 1.00. A higher gap reflects a lower degree of emotional well-being. The results of our study provide widespread confirmation of our hypothesis, with only two notable deviant cases (Vietnam and Kazakhstan), whose specific contexts are discussed later in this project.

1. Context & Professional Relevance

- **Technical Perspective**: We employed data analysis techniques (normalization, aggregation, and correlation analysis) on the World Happiness Report 2024 dataset to test the hypothesis that longevity is not linearly correlated with emotional well-being.
- **Economic Perspective**: Understanding this gap is crucial for public policy and the insurance/healthcare sectors. An increase in average life expectancy without a corresponding increase in emotional well-being leads to high social costs and unsustainable pressure on welfare systems.
- **Professional Interest**: The project integrated our multidisciplinary expertise to provide data interpretation models that move beyond GDP or simple life expectancy metrics.

2. Project Objectives

The primary objective is to quantify the **Health-Emotion Gap** globally, identifying countries in a ‘Silent Crisis’ (high health, low happiness). Our goals include: 1. Consolidating data from the 2024 World Happiness Report. 2. Normalizing health and affect parameters for direct comparability.

3. Team & Expertise

The group combines diverse backgrounds for a holistic view of the problem (in alphabetical order): - **Elia Bescotti**: Doctoral Degree in Political and Social Sciences; - **Manfred Meyer**: Master's Degree in Business Administration; - **Mai-Tam Tram**: Master of Science in Neurobiology.

Note: The methodology is largely new to the team, requiring intensive trial and error through research and statistical validation.

4. Interactions & Similar Projects

- **Expert Consultation**: Our primary point of reference and discussion has been Yaniv, our mentor. We reached out to other experts only as a means to fine-tune our data interpretation. Namely, we explored the possibility of accessing more detailed data for specific age groups, but we were not able to find a suitable partnership to access the data.
- **Benchmarking**: While studies on the 'Longevity Economy' exist, our project contributed by specifically isolating the relationship between positive/negative emotions (Affect Ratio) and physical health.

5. Data Understanding & Framework

- **Dataset**: World Happiness Report 2024 (Open Data).
- **Availability**: The data is freely available for research purposes.
- **Volume**: 2,363 initial observations with 11 core socio-economic indicators.

6. Relevance

To develop our measurement of reference, the Health-Emotion Gap, we identified that the most important data used in our analysis belongs to the 'Life Ladder', 'Healthy life expectancy at birth', 'Positive affect', and 'Negative affect' columns—all data connected with emotional and life-span aspects (for more information, see Appendix 1, World Happiness Report 2024, pp. 1 ff.).

We calculated the Health-Emotion Gap based on a linear transformation to eliminate unit differences between life expectancy and the affect ratio. On this basis, we mapped both metrics to a uniform 0–1 scale. For instance, a country with the lowest global life expectancy is assigned a 0.0, while the highest is assigned a 1.0. We applied the same procedure to the emotional affect ratio to compare these two dimensions directly. The final gap is defined as **Health - Emotion**, allowing us to isolate countries that possess a higher healthy life expectancy but suffer from a severe deficit in emotional happiness, offering a basis to confirm our hypothesis that longevity does not inherently guarantee daily emotional equilibrium.

7. Pre-processing and Feature Engineering

To ensure our comparison between healthy life expectancy and emotional joy was accurate, we performed three essential steps:

1. **Refining the Dataset (Cleaning & Interpolation)**: Upon inspecting the 2024 World Happiness Report, we discovered significant gaps in key indicators. To address this, we conducted thorough data cleaning. For countries with existing historical trends, we used **interpolation** (filling gaps with a country's mean values) to preserve as much data as possible. However, where missing values were too prevalent to allow for a reliable estimate (such as with the **State of Palestine**) we chose to remove the country entirely. This ensures our analysis remains statistically robust rather than speculative.
2. **Handling Special Cases**: We also identified that some entities with limited international recognition or sparse data reporting can distort global averages. For this reason, we analyzed certain cases (like **Taiwan**) separately later in the project to show how their specific data limitations might affect the broader 'Health-Emotion' scale.
3. **Creating a Level Playing Field (Normalization)**: Since health (measured in years) and emotions (measured as a ratio) use entirely different units, we 'normalized' them. By converting both to a uniform **0 to 1 scale**, we were able to directly compare a country's physical success against its daily emotional well-being.

This preparation allowed us to calculate the **Health-Emotion Gap** and identify where high life expectancy is failing to translate into daily happiness.

8. Visualizations and Statistics

Throughout the project, we have developed the following statistical analyses and visualizations. We published the most important ones in our PowerBI dashboard:

1. The Linearity Test (Scatter & Regression Plots): These are the first charts in your analysis. They visualize the raw relationship between Healthy Life Expectancy and Positive/Negative Affect, statistically confirming the 'cloud' of data points and weak correlation (0.22) that justifies the entire study.
2. Global & Regional Gap Trends (Line Plots): Located before the deep-dive statistics, these show the general upward trajectory of the 'Health-Emotion Gap' worldwide since 2006.
3. Preliminary Correlation Heatmap: This first heatmap provides a broad overview of how all indicators (GDP, Corruption, Social Support) relate to the Gap before focusing on the specific emotional disconnect.
4. The Medical Triumph (Regional Line Plot): A clear visualization of the successful global rise in healthy life expectancy across all ten macro-regions.
5. The Plot Twist (Advanced Heatmap): This is your second, more focused heatmap. It uses significance stars (***) to prove that while health and wealth correlate perfectly with 'rational' life evaluation (0.80), they decouple almost entirely from daily emotional 'affect' (0.19).
6. The Global Emotion Map (Quadrant Scatter Plot): A 3-year robust average that categorizes nations into archetypes, specifically isolating the 'Silent Crisis' (high health, low emotion).
7. Ranking the Silent Crisis (Bar Chart): A quantifiable ranking identifying the Middle East & North Africa and Western Europe as the regions where the gap is most severe.

8. Anatomy of a Crisis (Longitudinal Case Studies): Detailed historical 'micro-analyses' of Türkiye (the case with the highest Health-Emotion Gap discovered) and Vietnam and Kazakhstan (the two deviant cases) showing how economic or political shocks cause emotional well-being to crash even while biological longevity continues to climb.

9. Classification of the problem

Problem Statement and Visualization Type Our project addresses a complex storytelling and insight-communication problem: how to reveal the decoupling of biological longevity from emotional well-being within large-scale socio-economic datasets. While traditional metrics often assume a linear relationship between health and happiness, we aim to visualize the 'Health-Emotion Gap' to show that longer lifespans do not inherently equate to better emotional states. By presenting this divergence, we transform high-dimensional data from the World Happiness Report into a narrative that challenges existing policy assumptions regarding the 'Longevity Economy.'

Visualization Goals The primary goal is to enable **key pattern identification** and **interactive exploration** for researchers. We aim to move beyond static snapshots by using longitudinal case studies (like Türkiye) and archetype mapping (the scatterplot quadrants) to highlight the 'Silent Crisis.' Our visualizations are designed to isolate where wealth and health fail to translate into daily joy, ultimately opening the floor for the inclusion of exogenous factors (such as political stability or pandemic shocks) in national welfare strategies.

Performance Metrics and Evaluation Criteria The effectiveness of our visualizations and dashboard is assessed based on the clarity and interpretability of four specific KPIs: the **Affect Ratio**, the **Life Ladder**, **Healthy Life Expectancy at Birth**, and our custom-engineered **Health-Emotion Gap**. The success of our dashboard is measured by its ability to accurately distinguish cases of 'synchronized growth' (like Kazakhstan, where health and emotion move together) from 'temporal anomalies' (like Vietnam, where recent euphoria masks historical volatility), ensuring the user can accurately identify which countries are successfully bridging the gap versus those in crisis.

10. Dashboard design and optimization

Structure and Integration We developed a dynamic dashboard in PowerBI designed to bridge the gap between high-level executive summaries and detailed academic deep-dives. The interface is anchored by a global summary page featuring a central **World Map**, which serves as an intuitive entry point for non-specialist users. Surrounding this are our core analytical visualizations (the Regional Line Plots and the Global Emotion Map, i.e., the Scatterplot)integrated into a cohesive flow that allows users to move from global trends to specific regional hierarchies and finally to individual country narratives.

Interactive Features and User Experience To enhance exploration, we implemented a comprehensive suite of **slicers** that allow users to filter the entire dashboard by Year, Country, and Macro-Region. A key highlight of the user experience is the **interactive tooltip** integrated into the World Map; by hovering over any nation, the user triggers a pop-up visualization of the 'Anatomy of a Crisis' line plot for that specific country. This allows for a rapid, granular comparison of physical health versus emotional affect without leaving the main view. Furthermore, critical KPIs are displayed as standalone **call-out cards**, providing immediate clarity on the Affect Ratio and Health-Emotion Gap for the selected data subset.

Insights and Technical Optimizations The dashboard highlights patterns by color-coding regions consistently with our Python analysis, making the ‘Silent Crisis’ quadrants immediately identifiable. To ensure seamless performance and accurate data mapping, we performed a significant optimization: we manually synchronized the country name keys between our GeoJSON map file and the World Happiness Database. This was achieved by directly modifying the `.json` structure using Python in Colab (‘vibe coding’), ensuring that every country in our dataset correctly populates the map and triggers the appropriate longitudinal data call-outs without requiring a destructive overhaul of the primary database.

11. Interpretation of the results

Influence of Data Analysis on Design Our initial analysis of the visualizations highlighted the necessity of moving beyond static global views to more granular, comparative perspectives. By observing the wide variance in the ‘Health-Emotion Gap’ across different contexts, we decided to integrate **average values per region** into the dashboard. This addition allows users to benchmark specific countries against their geopolitical peers, providing a much-needed frame of reference that was missing in earlier iterations.

Clearer Communication through Interactivity To improve the clarity of data communication, we implemented dynamic **time and country filters** for the line plots. This interactive feature empowers users to conduct deep-dives into specific historical windows or individual national trajectories, transforming a complex dataset into a personalized narrative tool. We found that allowing the user to isolate data within a specific time range significantly reduced visual noise and clarified the impact of specific socio-economic shocks.

Accessibility and Interpretive Support Recognizing that the 4-quadrant scatterplot might be challenging for non-specialist audiences, we integrated the **World Map** as a geographic alternative. This provides an intuitive entry point that uses spatial familiarity to communicate the same underlying ‘Silent Crisis’ patterns. Furthermore, we optimized the dashboard’s usability by adding **executive notes for the heatmaps**. These concise explainers translate abstract correlation coefficients into actionable insights, ensuring that the ‘Plot Twist’—the decoupling of health and emotion—is immediately understood by stakeholders regardless of their statistical background.

12. Difficulties encountered through the project

Forecast: Mostly in line with the deadlines provided

Datasets: The datasets we used (World Happiness Report Datasets for 2024, aggregating later those from 2023, 2022, and 2021 for robustness) are publicly available. However, meaningful data filtered per age of the respondents was not available. This would have significantly increased the robustness of our results, since our focus is primarily on an older age category. Such a comparison can provide a fertile avenue for future research.

Technical/theoretical skills: The necessity of acquiring PowerBI skills for the visualization of the data was a challenge in terms of scheduling, as the deadline for the delivery of the exam fell within a sprint scheduled also for the submission of the final report (July 31). This aspect complicated a timely delivery of the report. However, Manfred was able to successfully develop draft interactive charts with Plotly as early as July 13, in order to have a clear interactive visualization available beforehand.

Relevance: Our research utilized a centralized database sufficient for core visualizations, though access to age-stratified survey data remained restricted behind institutional paywalls. Despite outreach efforts, these granular datasets were unavailable, marking a clear path for future research. Methodologically, heatmaps successfully illustrate the 'Plot Twist' decoupling, while line plots and bar plots provide longitudinal and regional context. The scatterplot archetypes nations via correlation-based KPIs, and country-specific case studies demonstrate that GDP and health metrics lack direct causation with happiness. These visuals reveal that emotional well-being often diverges from economic trends due to exogenous shocks and independent trajectories.

IT: Most of the work was conducted on Google Colab and PowerBI. Computational power and storage were not a relevant issue, although the necessity of a subscription to Microsoft 365 to use a shared version of the PowerBI dashboard made the interaction among the project members less smooth, as we had to rely on sharing copies through Google Drive. Finally, AI models (specifically Gemini) were helpful in streamlining the code, identifying and explaining errors, improving the visualizations (titles, labels, comments, colors), and cleaning the report language.

13. Report

Subdivision of tasks and duties:

Elia: *Main coding*: visualizations: correlation charts (fig. 1, 2), health-emotion gap line plots (fig. 3, 4), exploratory heatmap (fig. 5), plotting of Vietnam, Kazakhstan, and Taiwan cases (fig. 11, 12, 13); data interpretation, code cleaning, main report writing and editing; *Power BI dashboard*: data cleaning, executive overview and tooltip.

Mai: Project idea development; *main coding*: data cleaning, main calculations for indicators (Affect_Ratio, normalizations, Health-Emotion Gap); visualizations: 'The Medical Triumph' (fig. 6), 'The Plot-Twist' heatmap (fig. 7), scatter plot (fig. 8), ranking the silent crisis histogram (fig. 9), plotting of Turkey case (fig. 10), comments on Vietnam and Kazakhstan cases; data interpretation, report writing, and review; *Power BI dashboard*: introductory slides, dashboard review.

Manfred: *Main coding*: overall coding and report review, data export for Power BI; *Power BI Dashboard*: data arrangement, main dashbord bases creation, slicers application, DAX coding and translation from Python (heatmaps), executive notes.

Benchmarks:

We contextualized the **Health-Emotion Gap** against three primary benchmarks:

1. **Global Average**: Regional gaps in **MENA** and **Western Europe** exceed the ~0.32 global mean, indicating these regions underperform emotionally relative to their medical infrastructure.
2. **Theoretical Ideal (Utopia)**: Nearly 90% of healthy nations fall into the **'Silent Crisis'** quadrant, proving that the ideal of synchronized physical and emotional growth is globally elusive.

3. **Economic Wealth Assumptions**: While traditional models assume wealth drives well-being, our data reveals an 'Emotional Decoupling.' The correlation between health/wealth and daily joy is negligible (0.19), debunking the standard benchmark that economic success guarantees emotional happiness.

Achievements: * **Data Consolidation:** Successfully merged 2024 WHR data with historical trends (2006–2023) for 2,363 observations. * **Metric Engineering:** Normalised health and affect parameters to create the **Health-Emotion Gap**, enabling direct cross-unit comparison. * **Crisis Identification:** Quantified the 'Silent Crisis' archetype, identifying MENA and Western Europe as the primary regions of decoupling.

Partial Success & Limitations: * **Policy Application:** The 'National Equilibrium' remains a proof-of-concept. While case studies provide narrative evidence, specific policy prescriptions require age-stratified data currently restricted by institutional paywalls. Validating results against the 60+ demographic remains the primary target for future investigation once data access is secured.

14. Continuation of the project

The project offers several avenues for future research. First, as mentioned in the difficulties to access the data, a more granular overview of the responses to the WHR survey of the different age groups would be more than welcome. Indeed, focusing on respondents older than 60 would provide stronger evidence of our argument, or its rejection should that be the case.

Second, comparative and regional studies focusing on specific areas, economic zones, and human development standards, including external indicators where necessary, may help explain not only *if* a longer life expectancy does or does not translate into a higher degree of happiness, but also *why* this is the case. Particularly worthy of research could be a within-EU study, a CIS/Eurasian Economic Union study (where Kazakhstan represents a deviant case), or an ASEAN-focused study (where Vietnam is present).

Third, the dataset had problems and missing values for the states with limited international recognition included (namely Taiwan POC, State of Palestine, Kosovo, and Somaliland) as well as for Hong Kong. Incorporating survey research focusing on these entities, also not included in the datasets we used, could help refine our findings beyond the WHR data.

Cell 1: Importing Libraries and Loading Data

The dataset was loaded using `pd.read_csv()`. We imported all the libraries and functions that we considered useful for our analysis.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from scipy import stats
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')
```

Cell 2: Data Inspection & Structural Overview

We then proceed by extracting the dataset dimensions using `df.shape`. This step allows us to check for missing values (NaN) specifically within our core columns using `.isna().sum()`. Type verification ensures that our target features (Life Ladder, Healthy life expectancy at birth, etc.) are correctly interpreted by Python as numeric data types (`float64`) rather than strings (`object`).

```python
df = pd.read_csv('World-happiness-report-updated_2024.csv', encoding='latin1')

# 1. Check the shape of the dataset (rows, columns)
print("Dataset shape:", df.shape)

# 2. Check data types and overview of columns
print(df.info())

# 3. Count missing values (NaNs) for our core columns
# Focus on: 'Healthy life expectancy at birth', 'Positive affect', 'Negative
affect'
print(df[['Healthy life expectancy at birth', 'Positive affect', 'Negative
affect']].isna().sum())
```

```
Dataset shape: (2363, 11)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2363 entries, 0 to 2362
Data columns (total 11 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Country name                      2363 non-null   object
 1   year                              2363 non-null   int64
 2   Life Ladder                       2363 non-null   float64
 3   Log GDP per capita                2335 non-null   float64
 4   Social support                    2350 non-null   float64
 5   Healthy life expectancy at birth  2300 non-null   float64
 6   Freedom to make life choices      2327 non-null   float64
 7   Generosity                        2282 non-null   float64
 8   Perceptions of corruption         2238 non-null   float64
 9   Positive affect                   2339 non-null   float64
 10  Negative affect                   2347 non-null   float64
dtypes: float64(9), int64(1), object(1)
memory usage: 203.2+ KB
None
Healthy life expectancy at birth    63
Positive affect                     24
Negative affect                     16
dtype: int64
```

Cell 3: Missing Values

We then define a Python list of the four target columns that require cleaning or further processing. We used a `for` loop to iterate through the target columns. Inside the loop, we grouped the data by `Country name` and utilized `.apply()` with a lambda function to fill missing (NaN) values with each country's specific historical mean. Finally, we set `group_keys=False` to maintain the original dataframe alignment.

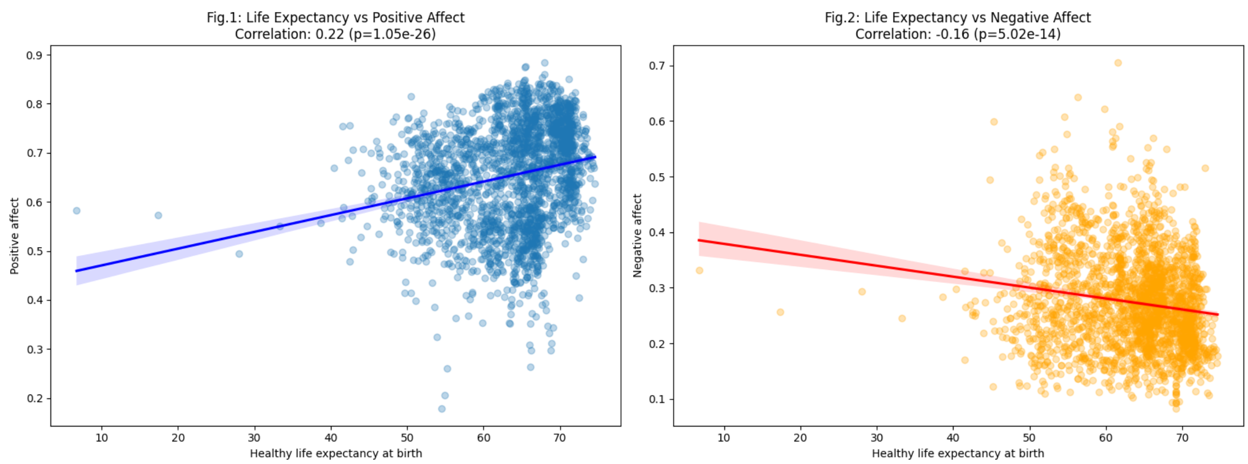
```
columns_to_impute = ['Life Ladder', 'Healthy life expectancy at birth', 'Positive
affect', 'Negative affect']

for col in columns_to_impute:
    # We use .apply() with group_keys=False to map the lambda function directly
    back to the original index
    df[col] = df.groupby('Country name', group_keys=False)[col].apply(lambda x:
x.fillna(x.mean()))

# Drop rows that still contain NaNs (countries with absolutely no data for these
columns)
df_clean = df.dropna(subset=columns_to_impute).copy()
```

Cell 4: Testing the linearity between healthy life expectancy and emotional well-being

We will now analyze the linear relationship between Healthy Life Expectancy and your emotional indicators. We calculate the correlation coefficients and provide scatter plots for both Positive and Negative Affect to visualize how they align with physical longevity.

```
from scipy.stats import pearsonr

# Prepare data by dropping missing values for the relevant columns
linearity_df = df_clean[['Healthy life expectancy at birth', 'Positive affect',
'Negative affect']].dropna()

# Calculate Pearson correlations
corr_pos, p_pos = pearsonr(linearity_df['Healthy life expectancy at birth'],
linearity_df['Positive affect'])
corr_neg, p_neg = pearsonr(linearity_df['Healthy life expectancy at birth'],
linearity_df['Negative affect'])

# Create the visualization
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Healthy Life Expectancy vs Positive Affect
sns.regplot(data=linearity_df, x='Healthy life expectancy at birth', y='Positive
affect',
            ax=axes[0], scatter_kws={'alpha':0.3}, line_kws={'color':'blue'})
axes[0].set_title(f'Fig.1: Life Expectancy vs Positive Affect\nCorrelation:
{corr_pos:.2f} (p={p_pos:.2e})')
```

```
# Healthy Life Expectancy vs Negative Affect
sns.regplot(data=linearity_df, x='Healthy life expectancy at birth', y='Negative
affect',
            ax=axes[1], scatter_kws={'alpha':0.3, 'color':'orange'},
line_kws={'color':'red'})
axes[1].set_title(f'Fig.2: Life Expectancy vs Negative Affect\nCorrelation:
{corr_neg:.2f} (p={p_neg:.2e})')

plt.tight_layout()
plt.show()

print("Correlation with Positive Affect:", corr_pos)
print("Correlation with Negative Affect:", corr_neg)
```




```
Correlation with Positive Affect: 0.21991304382092125
Correlation with Negative Affect: -0.15579310168757146
```

The statistical results are quite revealing. We found a weak positive correlation of 0.22 with Positive Affect and a weak negative correlation of -0.16 with Negative Affect.

Both results are statistically significant ($p < 0.05$), but the low coefficients confirm our hypothesis: biological longevity is a poor linear predictor of daily emotional states. While people are living longer, their physical health improvements are not translating proportionally into more joy or less stress.

In other words, the 'cloud' around the healthy life expectancy and emotional affect lines means that physical health is a necessary but not sufficient condition for happiness.

Cell 5: Creating the Affect_Ratio

We continue by creating a new metric, the **Affect_Ratio**, by dividing positive affect by negative affect, allowing us to present a unified value for emotional balance: - A value of 1 means that individuals have an equal ratio of positive to negative feelings. - A value of 3, for example, would show that positive emotions are three times higher than negative ones.

```
# Calculate the Affect_Ratio
df_clean['Affect_Ratio'] = df_clean['Positive affect'] / df_clean['Negative
affect']

# Quality check
print(df_clean['Affect_Ratio'].describe())
```

```
count    2311.000000
mean        2.678786
std         1.091924
min         0.294893
25%         1.900120
50%         2.526667
75%         3.292600
max         8.951807
Name: Affect_Ratio, dtype: float64
```

Cell 6: Normalization

To compare healthy life expectancy at birth to our affect ratio, we standardized both columns to range from 0 (worst data point) to 1 (best data point). We eliminated the unit differences between life expectancy (values spanning ~45–75 years) and affect ratio (values spanning ~0.5–4.0). We used the following formula:

`scaled value = (value - min) / (max - min)`

Example for life expectancy: - min = 45 years - max = 75 years - total range = 30 years - For a country with the worst life expectancy: `(45 - 45) / (75 - 45) = 0 / 30 = 0.0`. - For a country with the best life expectancy: `(75 - 45) / (75 - 45) = 30 / 30 = 1.0`.

Thus, the span is set from 0 to 1. We performed the same procedure for the `Affect_Ratio` to enable comparison. This calculation allows us to highlight the conflict in the data: a high life expectancy does not necessarily mean higher emotional health. We used **Health - Emotions** to show countries with a physical advantage but a deficit in emotional happiness.

```
# Scale 'Healthy life expectancy at birth'
min_health = df_clean['Healthy life expectancy at birth'].min()
max_health = df_clean['Healthy life expectancy at birth'].max()

df_clean['Health_Scaled'] = (df_clean['Healthy life expectancy at birth'] -
min_health) / (max_health - min_health)

# Scale 'Affect_Ratio'
min_affect = df_clean['Affect_Ratio'].min()
max_affect = df_clean['Affect_Ratio'].max()

df_clean['Affect_Ratio_Scaled'] = (df_clean['Affect_Ratio'] - min_affect) /
(max_affect - min_affect)
```

```python
# Calculate the gap (Health minus Emotion)
df_clean['Health_Emotion_Gap'] = df_clean['Health_Scaled'] -
df_clean['Affect_Ratio_Scaled']

print(df_clean[['Country name', 'year',
'Health_Emotion_Gap']].sort_values(by='Health_Emotion_Gap',
ascending=False).head(10))
```

```
     Country name  year  Health_Emotion_Gap
988        Israel  2023            0.902143
2152      Türkiye  2021            0.875043
2153      Türkiye  2022            0.858346
2151      Türkiye  2020            0.858144
1192      Lebanon  2021            0.856188
2154      Türkiye  2023            0.839179
1190      Lebanon  2019            0.835349
2150      Türkiye  2019            0.833804
777        Greece  2013            0.832989
1193      Lebanon  2022            0.830631
```

Understanding the Health-Emotion Gap

The **Health-Emotion Gap** is a simple score that tells us if a country's physical health (how long people live) matches its emotional health (how people actually feel day-to-day).

- **Value close to 0:** The country is in balance. Improvements in healthcare are perfectly reflected in the citizens' happiness.
- **High positive value (towards 1):** This signals a **Silent Crisis**. It means that while the country is excellent at keeping people alive and healthy, it is struggling to keep them happy. In these countries, a high gap value suggests a significant lack of joy, laughter, and positive daily emotions.
- **Negative value (towards -1):** This represents a paradoxical resilience. People report high levels of daily joy and emotional well-being despite having limited access to advanced healthcare or lower life expectancy.

```python
df_clean
```

	Country name	year	Life Ladder	Log GDP per capita	Social support	Healthy life expectancy at b
0	Afghanistan	2008	3.724	7.350	0.451	50.500
1	Afghanistan	2009	4.402	7.509	0.552	50.800
2	Afghanistan	2010	4.758	7.614	0.539	51.100
3	Afghanistan	2011	3.832	7.581	0.521	51.400
4	Afghanistan	2012	3.783	7.661	0.521	51.700
...	...	...	...	...	...	...

	Country name	year	Life Ladder	Log GDP per capita	Social support	Healthy life expectancy at b
2358	Zimbabwe	2019	2.694	7.698	0.759	53.100
2359	Zimbabwe	2020	3.160	7.596	0.717	53.575
2360	Zimbabwe	2021	3.155	7.657	0.685	54.050
2361	Zimbabwe	2022	3.296	7.670	0.666	54.525
2362	Zimbabwe	2023	3.572	7.679	0.694	55.000

Data Augmentation (Categorical Region Mapping)

After a first overview and study of the data, we realized that categorizing data solely by continent provides too broad a perspective for a nuanced analysis. For instance, baseline metrics for life expectancy and emotional well-being in Western Europe (e.g., Denmark, Switzerland) diverge significantly from those in Eastern Europe (e.g., Bulgaria, Albania). Moreover, we noted a similar structural discrepancy between Sub-Saharan Africa and North Africa, with the latter being culturally and economically more aligned with the Middle East. To account for these variances, we adopted the official standard used in global research and the World Happiness Report, stratifying our dataset into 10 distinct geopolitical macro-regions: we mapped these using the `Country name` column, used `.fillna('Other')` for unlisted countries, and assigned missing countries to their respective macro-regions.

```
region_map = {
    # --- WESTERN EUROPE ---
    'Austria': 'Western Europe', 'Belgium': 'Western Europe', 'Cyprus': 'Western
    Europe',
    'Denmark': 'Western Europe', 'Finland': 'Western Europe', 'France': 'Western
    Europe',
    'Germany': 'Western Europe', 'Greece': 'Western Europe', 'Iceland': 'Western
    Europe',
    'Ireland': 'Western Europe', 'Italy': 'Western Europe', 'Luxembourg':
    'Western Europe',
    'Malta': 'Western Europe', 'Netherlands': 'Western Europe', 'Norway':
    'Western Europe',
    'Portugal': 'Western Europe', 'Spain': 'Western Europe', 'Sweden': 'Western
    Europe',
    'Switzerland': 'Western Europe', 'United Kingdom': 'Western Europe',

    # --- CENTRAL AND EASTERN EUROPE ---
    'Albania': 'Central & Eastern Europe', 'Bosnia and Herzegovina': 'Central &
    Eastern Europe',
    'Bulgaria': 'Central & Eastern Europe', 'Croatia': 'Central & Eastern
    Europe',
    'Czechia': 'Central & Eastern Europe', 'Estonia': 'Central & Eastern Europe',
    'Hungary': 'Central & Eastern Europe', 'Kosovo': 'Central & Eastern Europe',
    'Latvia': 'Central & Eastern Europe', 'Lithuania': 'Central & Eastern
    Europe',
    'Montenegro': 'Central & Eastern Europe', 'North Macedonia': 'Central &
    Eastern Europe',
```

```
    'Poland': 'Central & Eastern Europe', 'Romania': 'Central & Eastern Europe',
    'Serbia': 'Central & Eastern Europe', 'Slovakia': 'Central & Eastern Europe',
    'Slovenia': 'Central & Eastern Europe',

    # --- COMMONWEALTH OF INDEPENDENT STATES (CIS) ---
    'Armenia': 'CIS', 'Azerbaijan': 'CIS', 'Belarus': 'CIS', 'Georgia': 'CIS',
    'Kazakhstan': 'CIS', 'Kyrgyzstan': 'CIS', 'Moldova': 'CIS', 'Russia': 'CIS',
    'Tajikistan': 'CIS', 'Turkmenistan': 'CIS', 'Ukraine': 'CIS', 'Uzbekistan':
    'CIS',

    # --- NORTH AMERICA AND ANZ ---
    'Australia': 'North America & ANZ', 'Canada': 'North America & ANZ',
    'New Zealand': 'North America & ANZ', 'United States': 'North America & ANZ',

    # --- LATIN AMERICA AND CARIBBEAN ---
    'Argentina': 'Latin America & Caribbean', 'Belize': 'Latin America &
    Caribbean',
    'Bolivia': 'Latin America & Caribbean', 'Brazil': 'Latin America &
    Caribbean',
    'Chile': 'Latin America & Caribbean', 'Colombia': 'Latin America &
    Caribbean',
    'Costa Rica': 'Latin America & Caribbean', 'Cuba': 'Latin America &
    Caribbean',
    'Dominican Republic': 'Latin America & Caribbean', 'Ecuador': 'Latin America
    & Caribbean',
    'El Salvador': 'Latin America & Caribbean', 'Guatemala': 'Latin America &
    Caribbean',
    'Guyana': 'Latin America & Caribbean', 'Haiti': 'Latin America & Caribbean',
    'Honduras': 'Latin America & Caribbean', 'Jamaica': 'Latin America &
    Caribbean',
    'Mexico': 'Latin America & Caribbean', 'Nicaragua': 'Latin America &
    Caribbean',
    'Panama': 'Latin America & Caribbean', 'Paraguay': 'Latin America &
    Caribbean',
    'Peru': 'Latin America & Caribbean', 'Suriname': 'Latin America & Caribbean',
    'Trinidad and Tobago': 'Latin America & Caribbean', 'Uruguay': 'Latin America
    & Caribbean',
    'Venezuela': 'Latin America & Caribbean',

    # --- MIDDLE EAST AND NORTH AFRICA (MENA) ---
    'Algeria': 'Middle East & North Africa', 'Bahrain': 'Middle East & North
    Africa',
    'Egypt': 'Middle East & North Africa', 'Iran': 'Middle East & North Africa',
    'Iraq': 'Middle East & North Africa', 'Israel': 'Middle East & North Africa',
    'Jordan': 'Middle East & North Africa', 'Kuwait': 'Middle East & North
    Africa',
    'Lebanon': 'Middle East & North Africa', 'Libya': 'Middle East & North
    Africa',
```

```
'Morocco': 'Middle East & North Africa', 'Oman': 'Middle East & North
Africa',
'Qatar': 'Middle East & North Africa', 'Saudi Arabia': 'Middle East & North
Africa',
'State of Palestine': 'Middle East & North Africa', 'Syria': 'Middle East &
North Africa',
'Tunisia': 'Middle East & North Africa', 'Turkey': 'Middle East & North
Africa',
'Türkiye': 'Middle East & North Africa', 'United Arab Emirates': 'Middle East
& North Africa',
'Yemen': 'Middle East & North Africa',

# --- SUB-SAHARAN AFRICA ---
'Angola': 'Sub-Saharan Africa', 'Benin': 'Sub-Saharan Africa', 'Botswana':
'Sub-Saharan Africa',
'Burkina Faso': 'Sub-Saharan Africa', 'Burundi': 'Sub-Saharan Africa',
'Cameroon': 'Sub-Saharan Africa',
'Central African Republic': 'Sub-Saharan Africa', 'Chad': 'Sub-Saharan
Africa',
'Comoros': 'Sub-Saharan Africa', 'Congo (Brazzaville)': 'Sub-Saharan Africa',
'Congo (Kinshasa)': 'Sub-Saharan Africa', 'Djibouti': 'Sub-Saharan Africa',
'Eswatini': 'Sub-Saharan Africa', 'Ethiopia': 'Sub-Saharan Africa', 'Gabon':
'Sub-Saharan Africa',
'Gambia': 'Sub-Saharan Africa', 'Ghana': 'Sub-Saharan Africa', 'Guinea':
'Sub-Saharan Africa',
'Ivory Coast': 'Sub-Saharan Africa', 'Kenya': 'Sub-Saharan Africa',
'Lesotho': 'Sub-Saharan Africa',
'Liberia': 'Sub-Saharan Africa', 'Madagascar': 'Sub-Saharan Africa',
'Malawi': 'Sub-Saharan Africa',
'Mali': 'Sub-Saharan Africa', 'Mauritania': 'Sub-Saharan Africa',
'Mauritius': 'Sub-Saharan Africa',
'Mozambique': 'Sub-Saharan Africa', 'Namibia': 'Sub-Saharan Africa', 'Niger':
'Sub-Saharan Africa',
'Nigeria': 'Sub-Saharan Africa', 'Rwanda': 'Sub-Saharan Africa', 'Senegal':
'Sub-Saharan Africa',
'Sierra Leone': 'Sub-Saharan Africa', 'Somalia': 'Sub-Saharan Africa', 'South
Africa': 'Sub-Saharan Africa',
'South Sudan': 'Sub-Saharan Africa', 'Sudan': 'Sub-Saharan Africa',
'Tanzania': 'Sub-Saharan Africa',
'Togo': 'Sub-Saharan Africa', 'Uganda': 'Sub-Saharan Africa', 'Zambia':
'Sub-Saharan Africa',
'Zimbabwe': 'Sub-Saharan Africa',

# --- SOUTH ASIA ---
'Afghanistan': 'South Asia', 'Bangladesh': 'South Asia', 'Bhutan': 'South
Asia',
'India': 'South Asia', 'Maldives': 'South Asia', 'Nepal': 'South Asia',
```

```python
    'Pakistan': 'South Asia', 'Sri Lanka': 'South Asia',

    # --- SOUTHEAST ASIA ---
    'Cambodia': 'Southeast Asia', 'Indonesia': 'Southeast Asia', 'Laos':
    'Southeast Asia',
    'Malaysia': 'Southeast Asia', 'Myanmar': 'Southeast Asia', 'Philippines':
    'Southeast Asia',
    'Singapore': 'Southeast Asia', 'Thailand': 'Southeast Asia', 'Vietnam':
    'Southeast Asia',

    # --- EAST ASIA ---
    'China': 'East Asia', 'Hong Kong S.A.R. of China': 'East Asia', 'Japan':
    'East Asia',
    'Mongolia': 'East Asia', 'South Korea': 'East Asia', 'Taiwan Province of
    China': 'East Asia'
}

df_clean['Region'] = df_clean['Country name'].map(region_map).fillna('Other')

print(df_clean['Region'].value_counts())
```

```
Region
Sub-Saharan Africa            519
Latin America & Caribbean     349
Western Europe                323
Central & Eastern Europe      259
Middle East & North Africa    253
CIS                           200
Southeast Asia                146
South Asia                    107
East Asia                      85
North America & ANZ            70
Name: count, dtype: int64
```

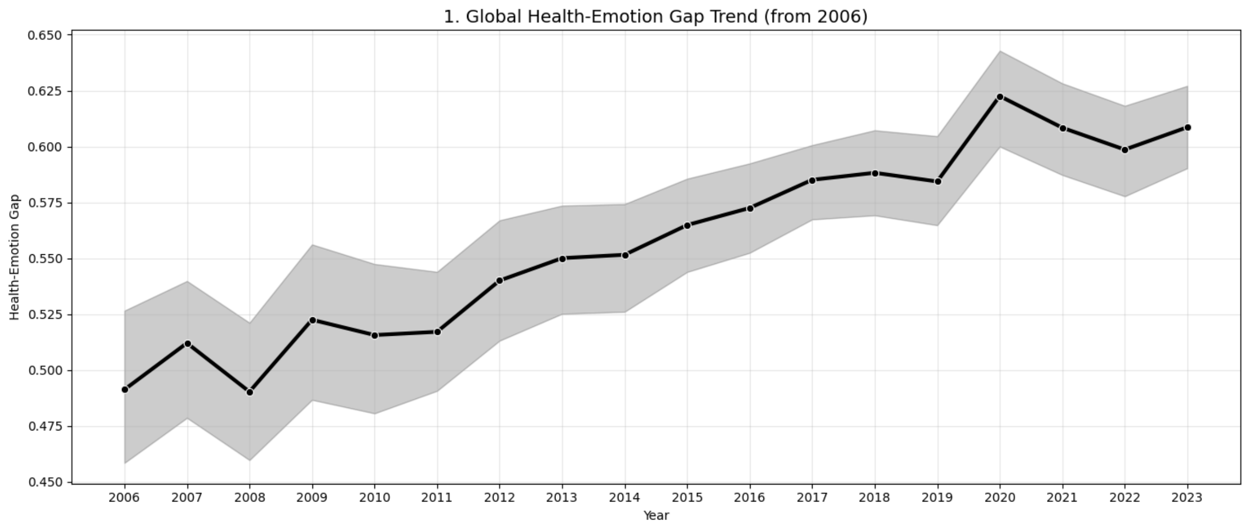
We noticed that the official WHR used external medical studies and global averages to estimate healthy life expectancy for places like Hong Kong and Kosovo. To maintain absolute data integrity and avoid speculative ratios, we made the conscious methodological choice to drop entities that lacked 100% of their primary health data.

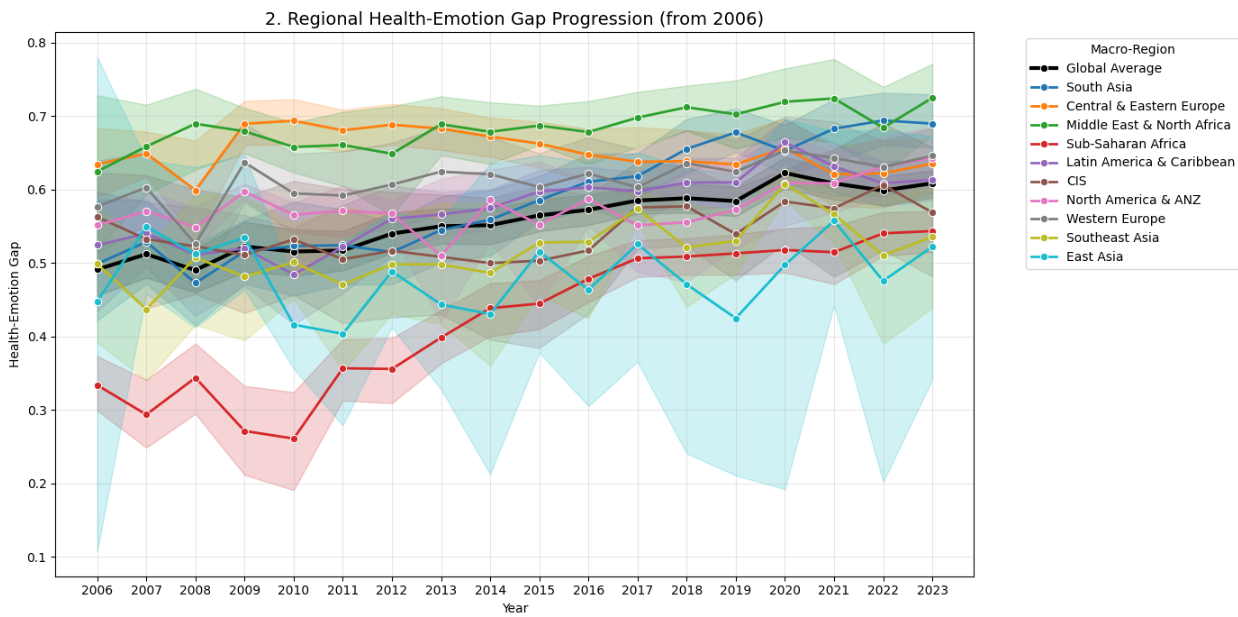
Note on Timeline Adjustment (2005 vs. 2006)

In our initial inspection of the dataset, we observed that data for the year 2005 is significantly sparse across several regions, most notably Sub-Saharan Africa, for the data necessary to calculate the Health-Emotion Gap (only the Happiness indicator was present). As such, including 2005 created visual distortions in our line plots, as the regional averages were based on an unrepresentative sample of countries. Henceforth, o ensure a robust and scientifically sound comparison of global and regional trends, we have calibrated our primary visualisations to start from 2006. This adjustment allows for a more stable baseline where data coverage is more consistent across all geopolitical macro-regions.

We now proceed to provide an overview of the trends for the gap between health and happiness over time, both worldwide and for each region.

```
# Filter the dataset to start specifically from 2006 for baseline stability
df_filtered = df_clean[df_clean['year'] >= 2006]

# Define the full range of years for the x-axis
years_range = sorted(df_filtered['year'].unique().astype(int))

# 1. Worldwide Trend Plot
plt.figure(figsize=(14, 6))
sns.lineplot(data=df_filtered, x='year', y='Health_Emotion_Gap', color = 'black',
linewidth=3, marker='o')
plt.xticks(years_range)
plt.xlabel("Year")
plt.ylabel("Health-Emotion Gap")
plt.title("1. Global Health-Emotion Gap Trend (from 2006)", fontsize=14)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# 2. Regional Trend Plot
plt.figure(figsize=(14, 7))
sns.lineplot(data=df_filtered, x='year', y='Health_Emotion_Gap', color = 'black',
linewidth=3, marker='o', label='Global Average')
sns.lineplot(data=df_filtered, x='year', y='Health_Emotion_Gap', hue='Region',
palette='tab10', marker='o', linewidth=2)
plt.xticks(years_range)
plt.xlabel("Year")
plt.ylabel("Health-Emotion Gap")
plt.title("2. Regional Health-Emotion Gap Progression (from 2006)", fontsize=14)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', title='Macro-Region')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



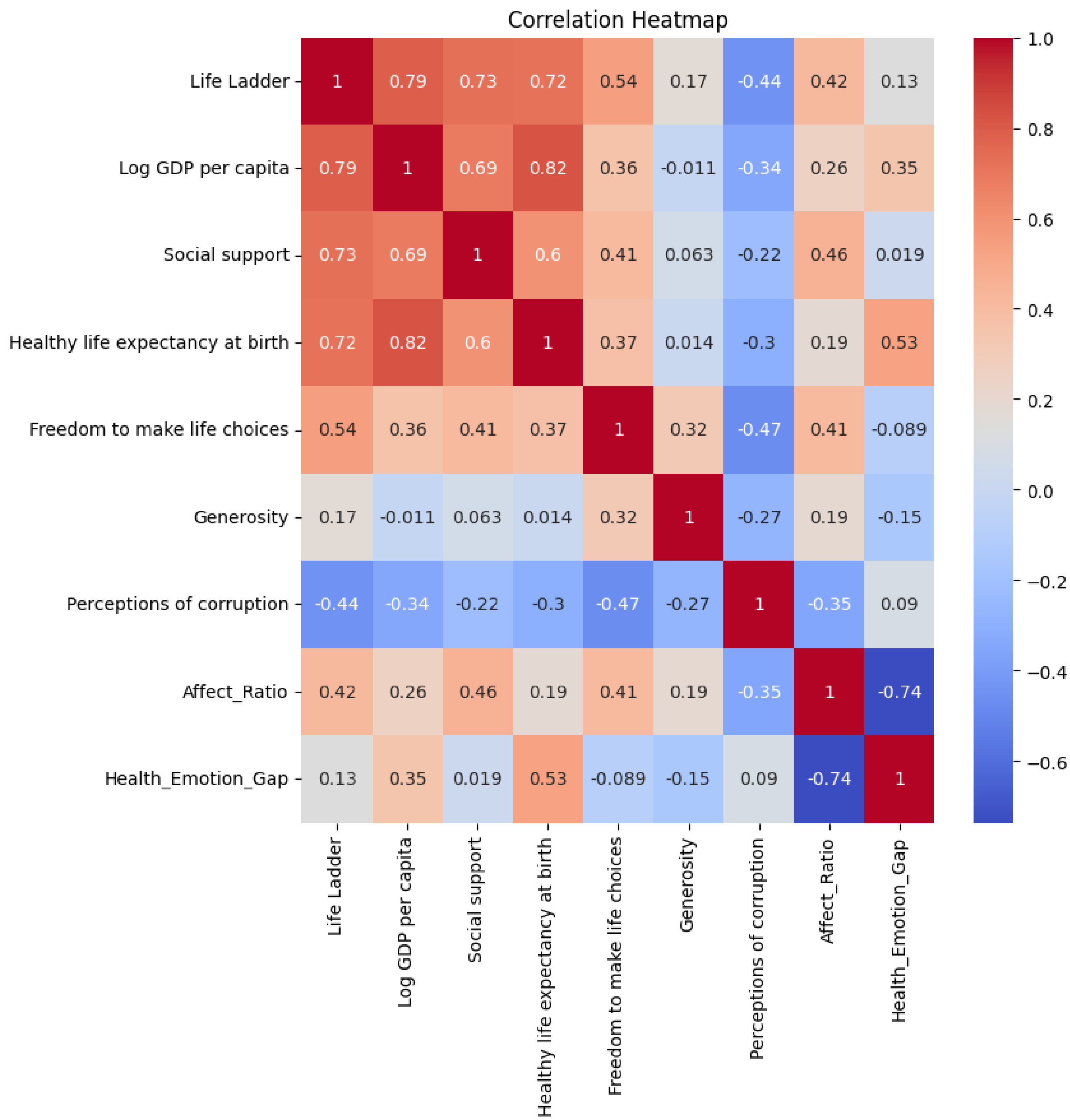


While the variation of indicators over time and within continents remains relevant, one can notice a general increase in the value of the Health-Emotion Gap over time. Sub-Saharan Africa is particularly low at the beginning of the graph, mostly due to a lower healthy life expectancy compared to other regions, while East Asia presents a rather low and varied value across time throughout the entire plot. After North America and ANZ, East Asia is the region with the fewest aggregated values (85), for which a substantial variation in a single country can significantly impact the performance of the average displayed.

To continue our analysis, we use a heatmap to understand which correlations are most relevant within the entire dataset. We limit our selection to indicators only, removing countries, regions, years, and scaled values.

```
numeric_df = df_clean.select_dtypes(include=np.number)
numeric_df =
numeric_df.drop(columns=['year',"Health_Scaled","Affect_Ratio_Scaled","Positive
affect","Negative affect"])
```

```
cor = numeric_df.corr()

plt.figure(figsize=(8, 8))
sns.heatmap(cor, annot=True, cmap="coolwarm") #we can use 'icefire', 'viridis',
'spectral', or 'vlag' if you prefer
plt.title("Correlation Heatmap")
plt.show()
```



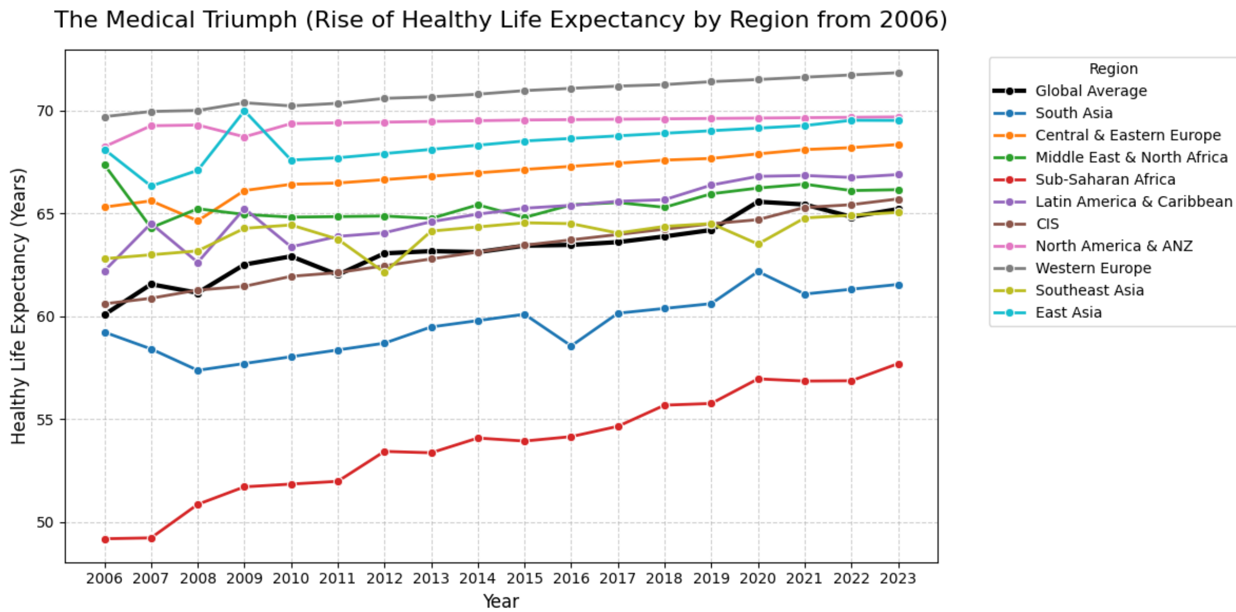
A Correlation Heatmap is a statistical visualization tool that displays how strongly different variables are mathematically related. The numbers inside the squares are correlation coefficients, ranging from -1.00 to +1.00. - **+1.00 (Dark Red)**: Perfect positive correlation. As variable A increases, variable B increases proportionally. (This is why the diagonal is always 1.00, as a variable correlates perfectly with itself.) - **0.00 (White/Light)**: No correlation. The variables move independently. - **-1.00 (Deep Blue)**: Perfect negative correlation. As variable A increases, variable B decreases.

The variable with the highest correlation with the Health-Emotion Gap is Healthy Life Expectancy at birth. This is unsurprising, as it was used as a base variable to calculate the gap. Given its importance, we provide a visualization of the trend subdivided by region over time in the following plot.

Besides health-related and emotion-related indicators, it appears that Log GDP per capita has a strong positive correlation with the Health-Emotion Gap, followed by Life Ladder, Perceptions of Corruption, and Social Support with lower values. In contrast, Generosity and Freedom to Make Life Choices present negative correlations. This is also expected, since lower happiness results in a higher Health-Emotion Gap. Finally, the very negative correlation with Affect Ratio also makes sense, because the Gap is meant to increase where the ratio is lower.

The next goal is to visually establish the global baseline of increasing biological longevity. To do so, we employed `seaborn.lineplot()` to map the historical trajectory of Healthy Life Expectancy at birth. The `hue` parameter was deployed to stratify the global trend by region. Also for the next visualization, the 'Medical Triumph' line plot, the data represents a historical time series starting from 2006. In this context, the programming library automatically calculates the annual mean for each region, meaning every point on the line represents the average healthy life expectancy of all countries within that specific macro-region for that given year.

```
plt.figure(figsize=(12, 6))

# Filter data from 2006
df_filtered = df_clean[df_clean['year'] >= 2006]
years_range = sorted(df_filtered['year'].unique().astype(int))
sns.lineplot(data=df_filtered, x='year', y='Healthy life expectancy at birth',
color = 'black', linewidth=3, marker='o', label='Global Average', errorbar=None)
sns.lineplot(data=df_filtered, x='year', y='Healthy life expectancy at birth',
hue='Region', marker='o', linewidth=2, errorbar=None)

plt.title('The Medical Triumph (Rise of Healthy Life Expectancy by Region from
2006)', fontsize=16, pad=15)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Healthy Life Expectancy (Years)', fontsize=12)

plt.legend(title='Region', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)

# Explicitly set ticks for every year in the range
plt.xticks(years_range)

plt.tight_layout()
plt.show()
```



The visualization displays 'The Medical Triumph' across all regions, confirming that humanity is systematically extending physical life. This sets the stage to question whether emotional well-being is keeping pace. This also applies to Sub-Saharan Africa and South Asia, lower in the chart compared to the others.

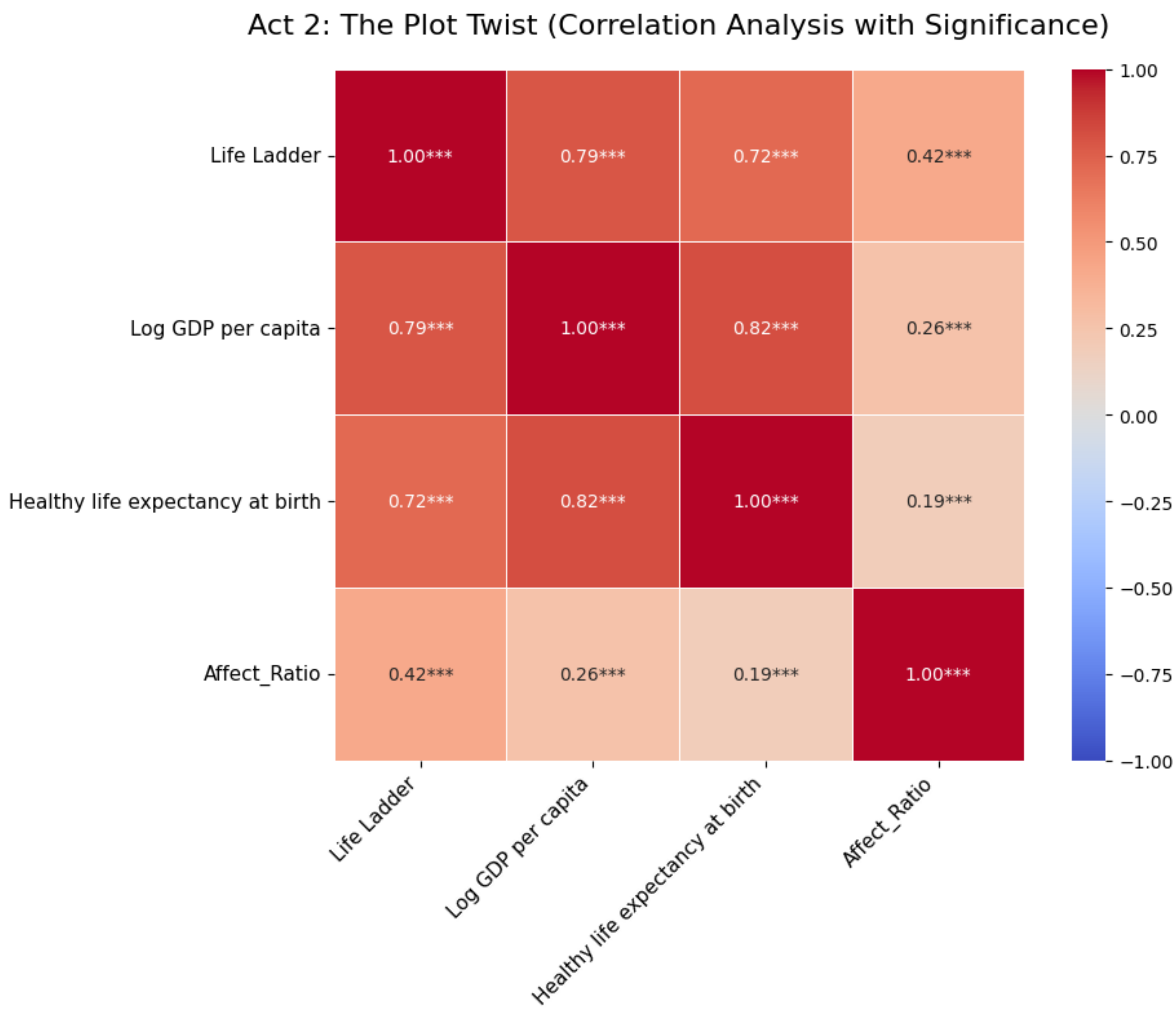
Our next step is to statistically test the assumption that physical longevity and economic wealth uniformly improve all dimensions of human well-being. While initially exploring the two linear correlations above, here we calculated a Pearson correlation matrix and visualized it via a Seaborn heatmap to provide a better picture of cross-correlations with other relevant indicators. We specifically contrasted the cognitive 'Life Ladder' metric with our engineered, emotion-based 'Affect_Ratio'.

The second visualization, the correlation heatmap, takes a pooled ordinary least squares approach. This means the correlation function evaluates the fundamental mathematical relationship between variables by treating every country-year combination as an independent observation, effectively pooling all available historical data to demonstrate the structural divergence between rational life evaluation and raw emotional affect.

```
# Define columns for correlation
columns_for_correlation = [
    'Life Ladder',
    'Log GDP per capita',
    'Healthy life expectancy at birth',
    'Affect_Ratio'
]

# Calculate the standard correlation matrix
correlation_matrix = df_clean[columns_for_correlation].corr()

# Create an empty copy for p-values
p_value_matrix = correlation_matrix.copy()
```

```
# Calculate the p-value for each pair of variables manually via SciPy
for col1 in columns_for_correlation:
    for col2 in columns_for_correlation:
        if col1 == col2:
            p_value_matrix.loc[col1, col2] = 0.0
        else:
            # Remove missing values for the specific pair to avoid calculation
            errors
            valid_data = df_clean[[col1, col2]].dropna()
            corr, p_val = pearsonr(valid_data[col1], valid_data[col2])
            p_value_matrix.loc[col1, col2] = p_val

# Create a matrix for the final annotations (Number + Stars)
annotation_matrix = correlation_matrix.copy().astype(str)

for i in range(len(columns_for_correlation)):
    for j in range(len(columns_for_correlation)):
        val = correlation_matrix.iloc[i, j]
        p_val = p_value_matrix.iloc[i, j]

        # Assign significance stars according to academic standards
        if p_val < 0.001:
            stars = '***'
        elif p_val < 0.01:
            stars = '**'
        elif p_val < 0.05:
            stars = '*'
        else:
            stars = ''

        annotation_matrix.iloc[i, j] = f"{val:.2f}{stars}"

# Set up the final heatmap
plt.figure(figsize=(10, 8))

sns.heatmap(
    correlation_matrix,
    annot=annotation_matrix,
    fmt='', # Format left empty as we pass strings from annotation_matrix
    cmap='coolwarm',
    vmin=-1,
    vmax=1,
    linewidths=0.5,
    square=True
)

plt.title('Act 2: The Plot Twist (Correlation Analysis with Significance)',
fontsize=16, pad=20)
```

```
plt.xticks(rotation=45, ha='right', fontsize=11)
plt.yticks(rotation=0, fontsize=11)
plt.tight_layout()
plt.show()
```



The Heatmap Explained

Our chart visually confirms the core hypothesis of the project. It can be divided into two logical sections: 1. **The Rational Cluster (Top Left)**: The correlations between Log GDP per capita (wealth), Healthy Life Expectancy (health), and the Life Ladder (rational life evaluation) are very strong (0.72 to 0.82). This shows that in wealthy, healthy nations, people rationally rate their lives highly. 2. **The Emotional Cluster (Bottom Right - The Twist)**: When looking at how physical health (Healthy Life Expectancy) correlates with actual daily emotions (Affect_Ratio), the value drops drastically to 0.19. Wealth (Log GDP per capita) also shows a weak correlation of 0.26 with daily emotions.

This heatmap provides statistical proof that while biological health and economic wealth drive cognitive life satisfaction (Life Ladder), they are poor predictors of whether people actually experience joy or suffer from stress in their daily lives (Affect Ratio).

The next step is to categorize countries into four distinct physical-emotional archetypes, specifically isolating those nations that exhibit high physical health alongside low daily emotional well-being—a phenomenon we define as the **Silent Crisis**.

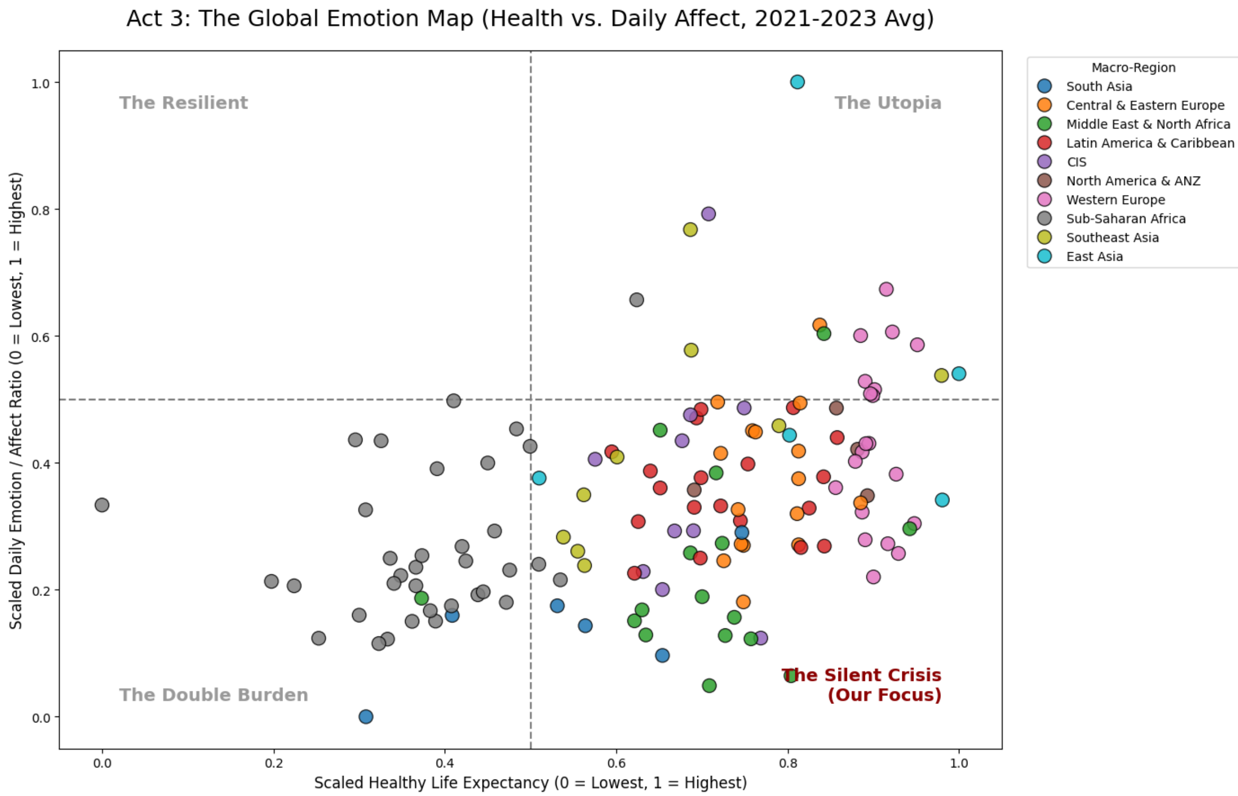
Regarding methodology, we aggregated data from the most recent three-year period to smooth out short-term statistical anomalies and provide a robust representation of the current global status quo (World Happiness Report, 2024).

Following this aggregation, we applied Min-Max scaling to both healthy life expectancy and the Affect Ratio, standardizing both metrics onto a uniform scale ranging from zero to one. We then generated a scatter plot partitioned into four quadrants by intersecting reference lines at exactly 0.5, with individual data points color-coded according to the ten macro-regions.

Each point represents the mean for one country over the years 2021–2023.

```
# Filter the dataset for the last three available years to align with official UN
methodology
df_recent = df_clean[df_clean['year'].isin([2021, 2022, 2023])]

# Calculate the 3-year robust average for each country
df_agg = df_recent.groupby(['Country name', 'Region'])[['Healthy life expectancy
at birth', 'Affect_Ratio']].mean().reset_index()

# Drop any countries that lack either health or emotion data in this specific
timeframe
df_agg = df_agg.dropna(subset=['Healthy life expectancy at birth',
'Affect_Ratio'])

# Apply Min-Max scaling to compress both metrics into a clean 0 to 1 range
health_min = df_agg['Healthy life expectancy at birth'].min()
health_max = df_agg['Healthy life expectancy at birth'].max()
df_agg['Health_Scaled'] = (df_agg['Healthy life expectancy at birth'] -
health_min) / (health_max - health_min)

affect_min = df_agg['Affect_Ratio'].min()
affect_max = df_agg['Affect_Ratio'].max()
df_agg['Affect_Ratio_Scaled'] = (df_agg['Affect_Ratio'] - affect_min) /
(affect_max - affect_min)

# Calculate the core metric: The Health-Emotion Gap
df_agg['Health_Emotion_Gap'] = df_agg['Health_Scaled'] -
df_agg['Affect_Ratio_Scaled']

# Initialize the plot dimensions
plt.figure(figsize=(14, 9))
```

```
# Generate the scatter plot with distinct regional colors
sns.scatterplot(
    data=df_agg,
    x='Health_Scaled',
    y='Affect_Ratio_Scaled',
    hue='Region',
    s=120,
    alpha=0.85,
    edgecolor='black'
)

# Draw the crosshairs at exactly 0.5 to create the four quadrants
plt.axvline(x=0.5, color='grey', linestyle='--', linewidth=1.5, zorder=0)
plt.axhline(y=0.5, color='grey', linestyle='--', linewidth=1.5, zorder=0)

# Add prominent text labels for the four narrative zones
plt.text(0.98, 0.98, 'The Utopia', fontsize=14, fontweight='bold', alpha=0.4,
ha='right', va='top')
plt.text(0.02, 0.02, 'The Double Burden', fontsize=14, fontweight='bold',
alpha=0.4, ha='left', va='bottom')
plt.text(0.02, 0.98, 'The Resilient', fontsize=14, fontweight='bold', alpha=0.4,
ha='left', va='top')
plt.text(0.98, 0.02, 'The Silent Crisis\n(Our Focus)', color='darkred',
fontsize=14, fontweight='bold', ha='right', va='bottom')

# Apply final formatting, titles, and layout adjustments
plt.title('Act 3: The Global Emotion Map (Health vs. Daily Affect, 2021-2023
Avg)', fontsize=18, pad=20)
plt.xlabel('Scaled Healthy Life Expectancy (0 = Lowest, 1 = Highest)',
fontsize=12)
plt.ylabel('Scaled Daily Emotion / Affect Ratio (0 = Lowest, 1 = Highest)',
fontsize=12)
plt.legend(title='Macro-Region', bbox_to_anchor=(1.02, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



Visual Structure of the Scatter Plot

- **The X-Axis (Health)**: Represents the 'Scaled Healthy Life Expectancy'. A value of 0 indicates the lowest global health outcomes, while 1 represents the highest medical longevity.
- **The Y-Axis (Emotion)**: Tracks the 'Scaled Daily Emotion / Affect Ratio'. A value of 0 means daily life is dominated by negative emotions (stress, sadness), while 1 indicates a high presence of positive emotions.
- **The Crosshair**: Dashed reference lines intersect at exactly 0.5 on both axes, mathematically dividing the map into four distinct narrative zones.
- **The Data Points**: Each dot represents a single country's robust three-year average, color-coded by macro-region.

Note: this map does not show the Health-Emotion gap

The Four Archetypes (Quadrants) Explained

1. **The Utopia (Top Right – High Health, High Emotion)**: This quadrant represents the ultimate societal ideal. Nations here offer citizens both a long, medically healthy life and a joyful daily existence. Notably, we found that only one significant outlier reaches this zone, highlighting how rare this balance is globally. However, on a deeper inspection, we identified that specific outlier (Taiwan) lacks significant data in terms of Healthy Life Expectancy (only three values available across 23 years) which strongly distorts its scaled results and the variables to it attached, as we show below.

2. **The Resilient (Top Left – Low Health, High Emotion)**: This zone characterizes societies that face medical challenges and low life expectancies yet maintain a positive emotional state. We found this quadrant to be empty in our global dataset.
3. **The Double Burden (Bottom Left – Low Health, Low Emotion)**: Countries in this area face a two-front battle: poor health infrastructure and a heavy emotional toll. We observed that this zone is heavily populated by Sub-Saharan African nations (grey dots).
4. **The Silent Crisis (Bottom Right – High Health, Low Emotion)**: This is the core focus of our research. Nations here have mastered biological longevity (far to the right) but remain trapped in the lower emotional spectrum. This proves that an excellent healthcare system does not automatically cure daily psychological stress.

Key Interpretations

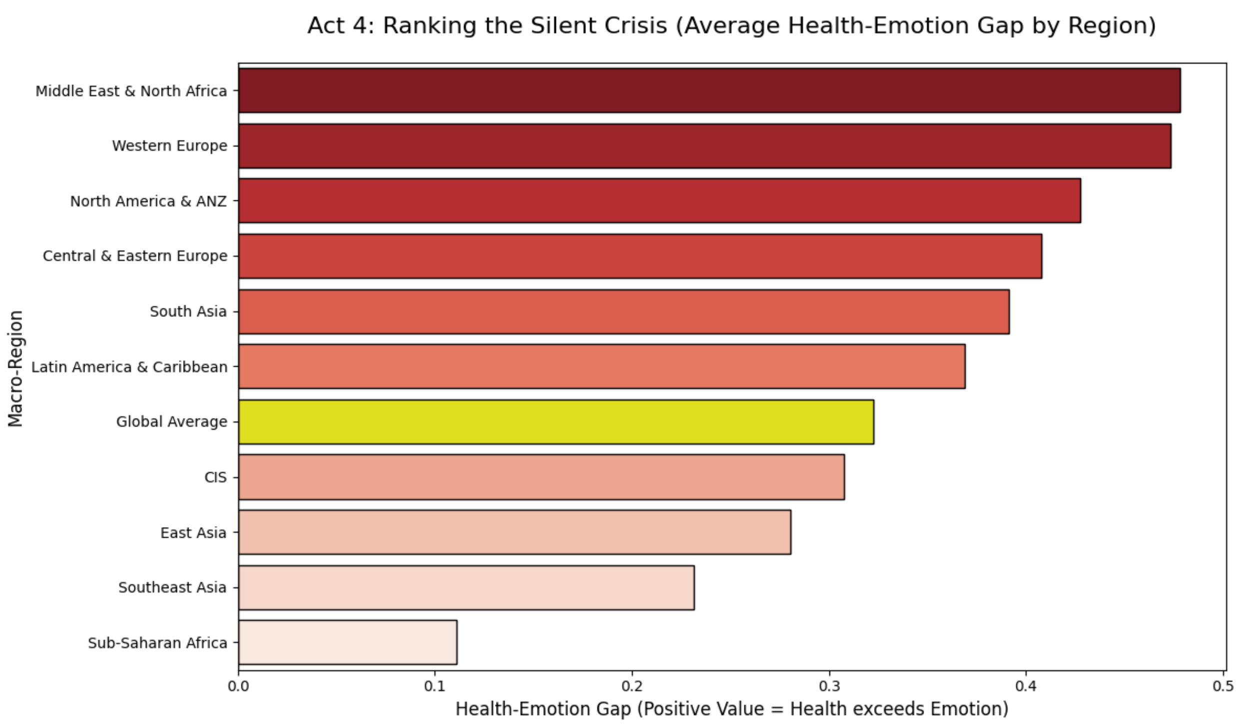
- **The Empty Upper Half**: The most striking revelation is the severe vertical asymmetry. We noted that almost the entire upper half of the system is devoid of countries, indicating that consistently high emotional well-being is globally elusive regardless of medical status.
- **The Density of the Crisis**: We observed that the 'Silent Crisis' quadrant is the most densely populated area, showing a massive agglomeration of developed and emerging nations.
- **Regional Clusters**: Within the crisis quadrant, we identified specific clusters featuring Western Europe (pink dots), Latin America (red dots), and Central/Eastern Europe (orange dots).

Consequently, the visualization provided compelling visual evidence that certain macro-regions are more subject suffer from a severe health-emotion gap, proving that biological longevity does not inherently guarantee daily emotional equilibrium.

Now, our objective is to quantify and rank the severity of the 'Silent Crisis' across all ten global macro-regions, translate individual data points from the preceding scatter plot into a clear, comparative hierarchy, and identify precisely which geographic areas suffer most from the discrepancy between high medical longevity and low daily emotional well-being. To do so, we adopted the standardized `Health_Scaled` and `Affect_Ratio_Scaled` metrics derived from the 2021–2023 average. For each country, we isolated the `Health_Emotion_Gap` by subtracting the scaled emotional affect from the scaled physical health. We subsequently grouped the dataset by macro-region, computed the mean gap for each cluster, and sorted the outcomes in descending order. We visualized the dataset via a horizontal bar chart, employing a sequential color palette to emphasize the magnitude of the discrepancy. We include also the global average as a parameter of comparison.

```
region_gap = df_agg.groupby('Region')['Health_Emotion_Gap'].mean().reset_index()
global_avg_val = df_agg['Health_Emotion_Gap'].mean()

# Creating a baseline row for the Global Average and concatenating it to the
regional data
global_row = pd.DataFrame({'Region': ['Global Average'], 'Health_Emotion_Gap':
[global_avg_val]})
plot_data = pd.concat([region_gap, global_row],
ignore_index=True).sort_values(by='Health_Emotion_Gap', ascending=False)

# We generate a base palette for regions and then include the global average
```

```
base_palette = sns.color_palette('Reds_r', n_colors=len(plot_data))
colors = ['yellow' if reg == 'Global Average' else base_palette[i]
          for i, reg in enumerate(plot_data['Region'])]

plt.figure(figsize=(12, 7))

sns.barplot(
    data=plot_data,
    y='Region',
    x='Health_Emotion_Gap',
    palette=colors,
    edgecolor='black')
plt.axvline(x=0, color='black', linestyle='-', linewidth=1)
plt.title('Act 4: Ranking the Silent Crisis (Average Health-Emotion Gap by
Region)', fontsize=16, pad=20)
plt.xlabel('Health-Emotion Gap (Positive Value = Health exceeds Emotion)',
fontsize=12)
plt.ylabel('Macro-Region', fontsize=12)

plt.tight_layout()
plt.show()
```



The visualization provides a definitive, quantifiable ranking of the macro-regions most severely affected by the physical-emotional paradox. Regions positioned at the top of the chart, characterized by the longest bars, demonstrate the most severe iterations of the 'Silent Crisis'. This statistically proves that in these specific areas, achievements in biological healthcare are disproportionate to the daily emotional equilibrium of their populations.

Building upon the findings of the regional analysis (Act 4), we algorithmically filtered our aggregated dataset to isolate the most severely affected macro-region: **Middle East & North Africa**.

Within this specific subset, all constituent nations were sorted in descending order based on the mathematical magnitude of their individual `Health_Emotion_Gap`.

```
mena_countries = df_agg[df_agg['Region'] == 'Middle East & North Africa']
mena_countries_sorted = mena_countries.sort_values(by='Health_Emotion_Gap',
ascending=False)

display(mena_countries_sorted[['Country name', 'Health_Emotion_Gap']].head(5))
```

	Country name	Health_Emotion_Gap
127	Türkiye	0.739494
70	Lebanon	0.659836
58	Israel	0.646492
63	Jordan	0.634581
55	Iran	0.599340

This computational sorting process objectively identifies the single nation with the most severe discrepancy in the region. The highest-ranking country is Türkiye and we subsequently select it as the ultimate focal point for our historical time-series analysis, serving as the prime global example of the physical-emotional paradox.

```
global_top_countries = df_agg.sort_values(by='Health_Emotion_Gap',
ascending=False)

display(global_top_countries[['Country name', 'Region',
'Health_Emotion_Gap']].head(5))
```

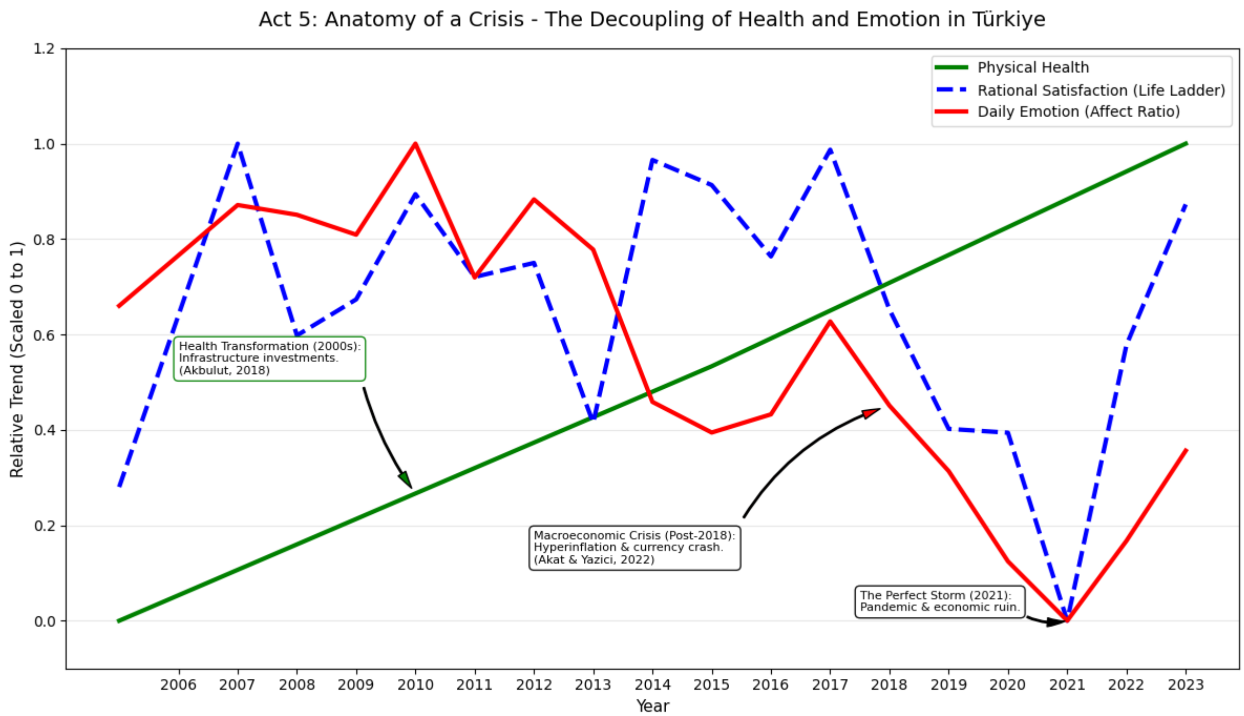
	Country name	Region	Health_Emotion_Gap
127	Türkiye	Middle East & North Africa	0.739494
80	Malta	Western Europe	0.680087
117	Spain	Western Europe	0.672200
70	Lebanon	Middle East & North Africa	0.659836
58	Israel	Middle East & North Africa	0.646492

At this point, we select Türkiye as our first case study, for which we want to provide a localized, longitudinal micro-analysis of a single representative nation currently situated within the 'Silent Crisis' quadrant, as well as to visually demonstrate the historical decoupling of biological longevity and cognitive life evaluation from daily emotional experiences over nearly two decades.

We isolated the historical time-series data (2006–present) for a specific target country. To enable a direct visual comparison of disparate metrics, we applied Min-Max scaling to the country's specific timeline. This transformation compressed physical health, the cognitive Life Ladder, and the emotional Affect Ratio onto an identical relative scale ranging from zero to one. As a result, we visualize the transformed variables using a multiple-line chart to track their trajectories over time.

```
target_country = 'Türkiye'
df_country = df_clean[df_clean['Country name'] == target_country].copy()

if df_country.empty:
    target_country = 'Turkey'
    df_country = df_clean[df_clean['Country name'] == target_country].copy()

df_country = df_country.sort_values('year')

# Apply Min-Max Scaling
metrics_to_scale = ['Healthy life expectancy at birth', 'Life Ladder',
'Affect_Ratio']

for col in metrics_to_scale:
    min_val = df_country[col].min()
    max_val = df_country[col].max()
    df_country[f'{col}_Trend'] = (df_country[col] - min_val) / (max_val -
min_val)

# Initialize the visualization figure
plt.figure(figsize=(12, 7))

# Plot the three distinct trend lines
sns.lineplot(data=df_country, x='year', y='Healthy life expectancy at
birth_Trend',
    label='Physical Health', color='green', linewidth=3)
sns.lineplot(data=df_country, x='year', y='Life Ladder_Trend', label='Rational
Satisfaction (Life Ladder)', color='blue', linewidth=3, linestyle='--')
sns.lineplot(data=df_country, x='year', y='Affect_Ratio_Trend', label='Daily
Emotion (Affect Ratio)', color='red', linewidth=3)

# Set comfortable y-axis limits
plt.ylim(-0.1, 1.2)

# Safe coordinate extraction
def get_trend_y(df, year_val, col_name):
    subset = df[df['year'] == year_val]
    if not subset.empty:
        return subset[col_name].values[0]
    else:
        return df.iloc[(df['year'] -
        year_val).abs().argsort()[:1]][col_name].values[0]
```

```
y_health_2010 = get_trend_y(df_country, 2010, 'Healthy life expectancy at
birth_Trend')
y_emotion_2018 = get_trend_y(df_country, 2018, 'Affect_Ratio_Trend')

# 1st annotation: Health Transformation
plt.annotate('Health Transformation (2000s):\nInfrastructure
investments.\n(Akbulut, 2018)',
    xy=(2010, y_health_2010),
    xytext=(2006, y_health_2010 + 0.25),
    arrowprops=dict(facecolor='green', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=0.1"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='green',
alpha=0.9))

# 2nd annotation: Macroeconomic Stress
plt.annotate('Macroeconomic Crisis (Post-2018):\nHyperinflation & currency
crash.\n(Akat & Yazici, 2022)',
    xy=(2018, y_emotion_2018),
    xytext=(2012, 0.12),
    arrowprops=dict(facecolor='red', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=-0.2"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='black',
alpha=0.9))

# 3rd annotation: 2021 Nadir
plt.annotate('The Perfect Storm (2021):\nPandemic & economic ruin.',
    xy=(2021, 0.0),
    xytext=(2017.5, 0.02),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=0.2"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='black',
alpha=0.9))


plt.title(f'Act 5: Anatomy of a Crisis - The Decoupling of Health and Emotion in
{target_country}', fontsize=14, pad=15)
plt.xlabel('Year', fontsize=11)
plt.ylabel('Relative Trend (Scaled 0 to 1)', fontsize=11)
plt.xticks(years_range)
plt.legend(loc='upper right')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



The Deep Dive Explanation

1. The Steady Rise of the Green Line (Physical Health) - Systemic Healthcare Reforms: In the early 2000s, Turkey implemented the 'Health Transformation Program,' overhauling the medical infrastructure. - **Medical Infrastructure**: Massive investments led to universal health coverage, modern hospitals, and a steep decline in infant mortality. - **The Biological Outcome**: These structural improvements mechanically drove up life expectancy and physical longevity, independently of the population's psychological state.

2. The Dramatic Collapse of the Red Line (Daily Emotion) - Chronic Macroeconomic Stress: Beginning around 2013 and accelerating post-2018, Turkey suffered a severe currency crisis. Hyperinflation and the collapse of the Lira obliterated purchasing power. This constant financial anxiety translates directly into high allostatic load. - **Societal and Geopolitical Instability**: The aftermath of the 2016 coup attempt, political polarization, and the Syrian refugee crisis created a pervasive climate of uncertainty. - **The 2021 Nadir**: The collision of the pandemic with peaking hyperinflation created a perfect storm. The simultaneous threats to physical survival and financial stability drove daily positive affect to the absolute bottom of the index.

3. The Core Takeaway: Modern medical infrastructure is effective at sustaining biological life and extending longevity. However, it is fundamentally incapable of buffering a population against chronic socio-economic stressors that devastate daily emotional well-being.

The visualization functions as a national case study, proving our central hypothesis on a micro-level. The trend lines reveal a structural divergence: while physical health metrics and rational life satisfaction remain stable or show upward trajectories, the trend for daily emotional affect stagnates or declines. This provides concrete proof that modern healthcare achievements are failing to translate into immediate psychological relief or joy in everyday life.

Yet, Türkiye represents a value particularly biased toward supporting our hypothesis. We therefore proceed by taking into consideration some countries located in the 'Utopia' quadrant—specifically those with the lowest gap. To do so, we order countries with a scaled life expectancy higher than the average by their Health-Emotion Gap in ascending order.

```
global_bottom_countries = df_agg.sort_values(by='Health_Emotion_Gap',
ascending=True)
healthy_global_bottom_countries =
global_bottom_countries[global_bottom_countries['Health_Scaled'] >
global_bottom_countries['Health_Scaled'].mean()]
selected_columns = healthy_global_bottom_countries[['Country name', 'Healthy life
expectancy at birth', 'Affect_Ratio', 'Health_Emotion_Gap']]
selected_columns.head(10)
#
```

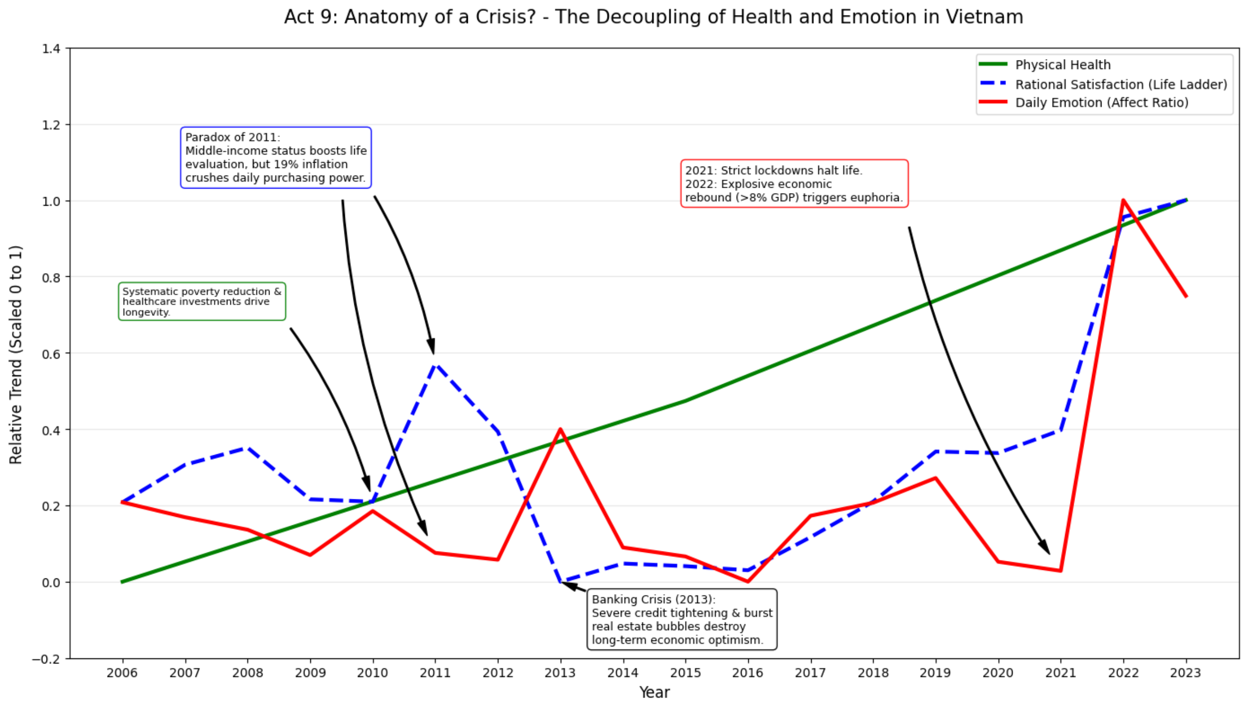
	Country name	Healthy life expectancy at birth	Affect_Ratio	Health_Emotion_Gap
121	Taiwan Province of China	69.1400	6.569624	-0.188350
64	Kazakhstan	66.2000	5.288349	-0.084246
136	Vietnam	65.6000	5.136111	-0.080727
78	Malaysia	65.6250	3.965471	0.110103
134	Uzbekistan	65.6000	3.337271	0.211151
101	Paraguay	65.9500	3.390138	0.214930
37	Estonia	69.8750	4.209588	0.220537
69	Latvia	66.5000	3.461941	0.222696
83	Mexico	65.8000	3.307121	0.223104
66	Kuwait	70.0125	4.126254	0.238913

The table returns only three negative values for the Health-Emotion Gap: Taiwan POC, Kazakhstan, and Vietnam. These three cases seem to contradict our hypothesis. However, on a deeper inspection, Taiwan POC presents only a limited sample of data in the database, with indicators for only three years out of 23 in terms of (scaled) healthy life expectancy that strongly distort the analysis and the outcome in terms of Health-Emotion Gap. While this does not rule out that the Taiwanese model or situation may be indeed the one with the lowest gap in the world, we have also identified two more consistent cases for our analysis also landing in the "Utopia" quadrant: Vietnam and Kazakhstan. We will proceed by studying them.

```
#Adopting the same steps used to filter and comment Türkiye
target_country = 'Vietnam'
df_country = df_clean[df_clean['Country name'] == target_country].copy()
df_country = df_country.sort_values('year')

metrics_to_scale = ['Healthy life expectancy at birth', 'Life Ladder',
'Affect_Ratio']
for col in metrics_to_scale:
    min_val = df_country[col].min()
```

```
    max_val = df_country[col].max()
    df_country[f'{col}_Trend'] = (df_country[col] - min_val) / (max_val -
min_val)

plt.figure(figsize=(14, 8))

sns.lineplot(data=df_country, x='year', y='Healthy life expectancy at
birth_Trend', label='Physical Health', color='green', linewidth=3)
sns.lineplot(data=df_country, x='year', y='Life Ladder_Trend', label='Rational
Satisfaction (Life Ladder)', color='blue', linewidth=3, linestyle='--')
sns.lineplot(data=df_country, x='year', y='Affect_Ratio_Trend', label='Daily
Emotion (Affect Ratio)', color='red', linewidth=3)
plt.ylim(-0.2, 1.4)

def get_trend_y(df, year_val, col_name):
    subset = df[df['year'] == year_val]
    if not subset.empty: return subset[col_name].values[0]
    else: return df.iloc[(df['year'] -
    year_val).abs().argsort()[:1]][col_name].values[0]

y_ladder_2011 = get_trend_y(df_country, 2011, 'Life Ladder_Trend')
y_emotion_2011 = get_trend_y(df_country, 2011, 'Affect_Ratio_Trend')
y_ladder_2013 = get_trend_y(df_country, 2013, 'Life Ladder_Trend')
y_emotion_2021 = get_trend_y(df_country, 2021, 'Affect_Ratio_Trend')

# adding notes to the graph
plt.annotate(    'Systematic poverty reduction &\nhealthcare investments
drive\nlongevity.',
    xy=(2010, get_trend_y(df_country, 2010, 'Healthy life expectancy at
birth_Trend')),
    xytext=(2006, 0.7),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=-0.1"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='green',
alpha=0.9))

plt.annotate('Paradox of 2011:\nMiddle-income status boosts life\nevaluation, but
19% inflation\ncrushes daily purchasing power.',
    xy=(2011, y_ladder_2011),
    xytext=(2007, 1.05),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=-0.1"),
    fontsize=9, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='blue',
alpha=0.9))

plt.annotate('',
    xy=(2011, y_emotion_2011),
```

```
    xytext=(2009.5, 1.05),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=0.1"))

plt.annotate('Banking Crisis (2013):\nSevere credit tightening & burst\nreal
estate bubbles destroy\nlong-term economic optimism.',
    xy=(2013, y_ladder_2013),
    xytext=(2013.5, -0.16),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=0"),
    fontsize=9, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='black',
alpha=0.9))

plt.annotate(    '2021: Strict lockdowns halt life.\n2022: Explosive
economic\nrebound (>8% GDP) triggers euphoria.',
    xy=(2021, y_emotion_2021),
    xytext=(2015, 1),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=0.1"),
    fontsize=9, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='red',
alpha=0.9))

plt.title('Act 9: Anatomy of a Crisis? - The Decoupling of Health and Emotion in
Vietnam', fontsize=15, pad=20)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Relative Trend (Scaled 0 to 1)', fontsize=12)
plt.xticks(df_country['year'].astype(int))
plt.legend(loc='upper right')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



The historical data for Vietnam reveals a highly volatile decoupling of physical health and emotional well-being, driven by a series of profound macroeconomic and social shocks. The steady rise in physical longevity is the direct biological result of long-term programs aimed at poverty reduction and systematic state investments in the healthcare system, which consistently drive up life expectancy independently of short-term crises. However, the psychological metrics reflect a much more turbulent reality. In 2011, Vietnam officially achieved middle-income country status, which initially sparked immense societal optimism and drove the cognitive Life Ladder evaluation sharply upward. Paradoxically, this same year saw a devastating inflation rate of nearly 19%, which severely reduced daily purchasing power and caused a sharp crash in daily positive emotions. The government's subsequent efforts to combat this extreme inflation through drastic interest rate hikes and strict credit tightening ultimately burst a domestic real estate bubble. This led to a severe banking crisis between 2013 and 2016, characterized by high rates of non-performing loans and corporate insolvencies, which dampened long-term economic optimism and caused the Life Ladder to decline. More recently, the catastrophic drop in daily emotions in 2021 marks the devastating impact of the COVID-19 Delta wave. Vietnam imposed some of the strictest lockdowns globally, specifically in economic hubs like Ho Chi Minh City, closing factories and isolating millions, bringing public life to a standstill. This emotional nadir was immediately followed by an explosive rebound in 2022. This spike reflects the societal and economic reopening, highlighted by GDP growth of over 8%, where the return to social normalcy and rapid recovery triggered widespread euphoria.

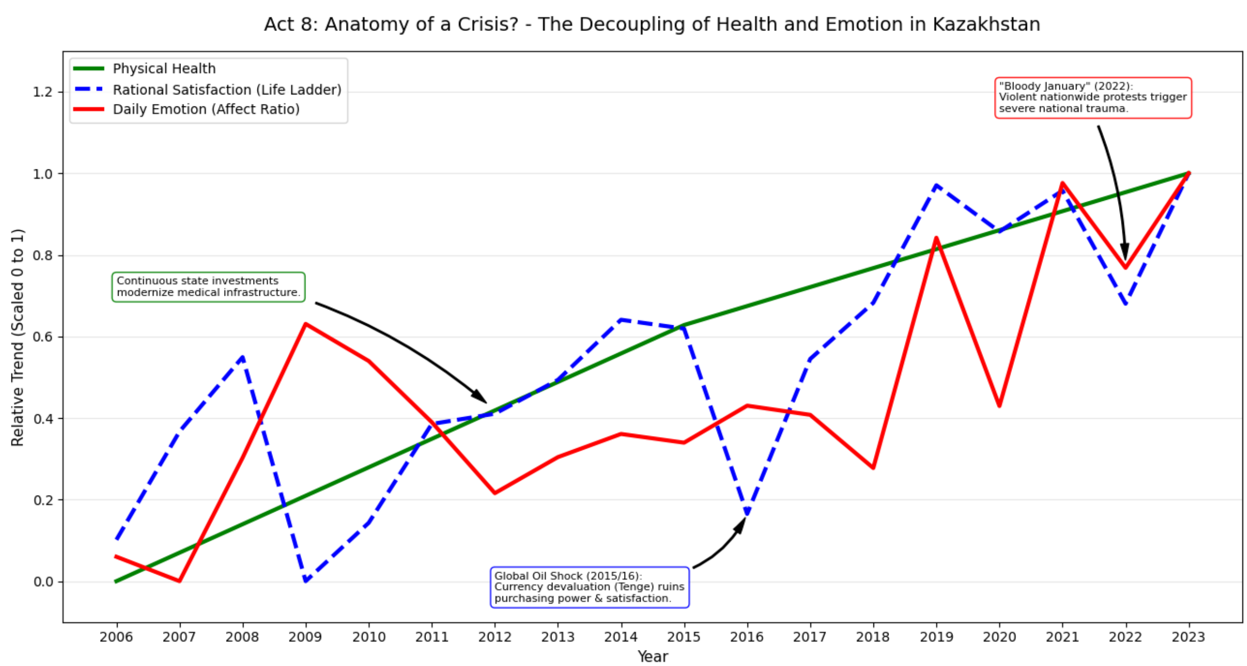
To sum up, the graph shows a peak in rational and emotional satisfaction right at the end of the graph, around the most recent years, which are those captured by our scatterplot. Thus, Vietnam's presence in the 'Utopia' quadrant seems to be more a recent exception than its historical trend.

We now proceed with studying the second deviant case of our dataset, that is Kazakhstan.

```
# We follow the same steps as above
target_country = 'Kazakhstan'
```

```
df_country = df_clean[df_clean['Country name'] == target_country].copy()
df_country = df_country.sort_values('year')

metrics_to_scale = ['Healthy life expectancy at birth', 'Life Ladder',
'Affect_Ratio']
for col in metrics_to_scale:
    min_val = df_country[col].min()
    max_val = df_country[col].max()
    df_country[f'{col}_Trend'] = (df_country[col] - min_val) / (max_val -
min_val)

plt.figure(figsize=(13, 7))
sns.lineplot(data=df_country, x='year', y='Healthy life expectancy at
birth_Trend', label='Physical Health', color='green', linewidth=3)
sns.lineplot(data=df_country, x='year', y='Life Ladder_Trend', label='Rational
Satisfaction (Life Ladder)', color='blue', linewidth=3, linestyle='--')
sns.lineplot(data=df_country, x='year', y='Affect_Ratio_Trend', label='Daily
Emotion (Affect Ratio)', color='red', linewidth=3)

plt.ylim(-0.1, 1.3)


def get_trend_y(df, year_val, col_name):
    subset = df[df['year'] == year_val]
    if not subset.empty: return subset[col_name].values[0]
    else: return df.iloc[(df['year'] -
    year_val).abs().argsort()[:1]][col_name].values[0]

# Annotations
plt.annotate('Continuous state investments\nmodernize medical infrastructure.',
    xy=(2012, get_trend_y(df_country, 2012, 'Healthy life expectancy at
birth_Trend')),
    xytext=(2006, 0.7),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=-0.1"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='green',
alpha=0.9))

plt.annotate('Global Oil Shock (2015/16):\nCurrency devaluation (Tenge)
ruins\npurchasing power & satisfaction.',
    xy=(2016, get_trend_y(df_country, 2016, 'Life Ladder_Trend')),
    xytext=(2012, -0.05),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=0.2"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='blue',
alpha=0.9))
```

```
plt.annotate('"Bloody January" (2022):\nViolent nationwide protests
trigger\nsevere national trauma.',
    xy=(2022, get_trend_y(df_country, 2022, 'Affect_Ratio_Trend')),
    xytext=(2020, 1.15),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6,
connectionstyle="arc3,rad=-0.1"),
    fontsize=8, bbox=dict(boxstyle='round,pad=0.4', fc='white', ec='red',
alpha=0.9))

plt.title('Act 8: Anatomy of a Crisis? - The Decoupling of Health and Emotion in
Kazakhstan', fontsize=14, pad=15)
plt.xlabel('Year', fontsize=11)
plt.ylabel('Relative Trend (Scaled 0 to 1)', fontsize=11)
plt.xticks(df_country['year'].astype(int))
plt.legend(loc='upper left')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



The Paradox of Kazakhstan: Resilience Amidst Instability

As already shown, unlike the majority of nations, Kazakhstan exhibits a negative Health-Emotion Gap (-0.084), meaning its daily emotional well-being is statistically higher than its healthy life expectancy would predict. While the consistent upward trend in physical health shows that the population benefits from state investments in medical infrastructure, the emotional data reveals a unique equilibrium. Historically, the data reflects deep structural shocks. The decline in rational satisfaction around 2015 and 2016 was triggered by a global oil price shock, forcing a devaluation of the Tenge and generating chronic financial stress. Similarly, the 2022 ‘Bloody January’ crisis created a profound national trauma that caused an abrupt crash in life satisfaction. However, the core narrative for Kazakhstan is one of avoiding the ‘Silent Crisis.’ Even with these shocks, Kazakhstan maintains an exceptionally high frequency of positive daily emotions. While the rest of the world sees a decoupling where biological life extends but joy stagnates, Kazakhstan defies this trend, keeping its emotional equilibrium ahead of its physical health milestones. This suggests that the population maintains a level of daily joy and resilience that persists despite severe socio-political devastation, showing a positive trend not only in terms of life expectancy, but also rational and emotional satisfaction. At this point, we need to confront how the indicators we identified in our Plot Twist heatmap apply to Kazakhstan.

```
df_kazakhstan = df_clean[df_clean['Country name'] ==
'Kazakhstan'].dropna(subset=['Affect_Ratio', 'Life Ladder', 'Healthy life
expectancy at birth', 'Log GDP per capita'])

corr_cols = ['Affect_Ratio', 'Life Ladder', 'Healthy life expectancy at birth',
'Log GDP per capita']

kazakhstan_corr = df_kazakhstan[corr_cols].corr()

print("Correlations for Kazakhstan:")
display(kazakhstan_corr[['Affect_Ratio']])
```

```
Correlations for Kazakhstan:
```

	Affect_Ratio
Affect_Ratio	1.000000
Life Ladder	0.545621
Healthy life expectancy at birth	0.691010
Log GDP per capita	0.592861

Interpreting Kazakhstan vis-à-vis the hypothesis: the exception confirming the rule

Based on the data we have analyzed, the case of Kazakhstan does not so much reject our hypothesis as it provides a critical ‘deviant case’ that defines its boundaries. Indeed, from a technical perspective, Kazakhstan contradicts the global trend of the ‘Silent Crisis.’ Relative to the global scale, its emotional well-being is higher than its biological longevity would predict.

The results show a moderately strong positive relationship between the `Affect_Ratio` and the other key variables:

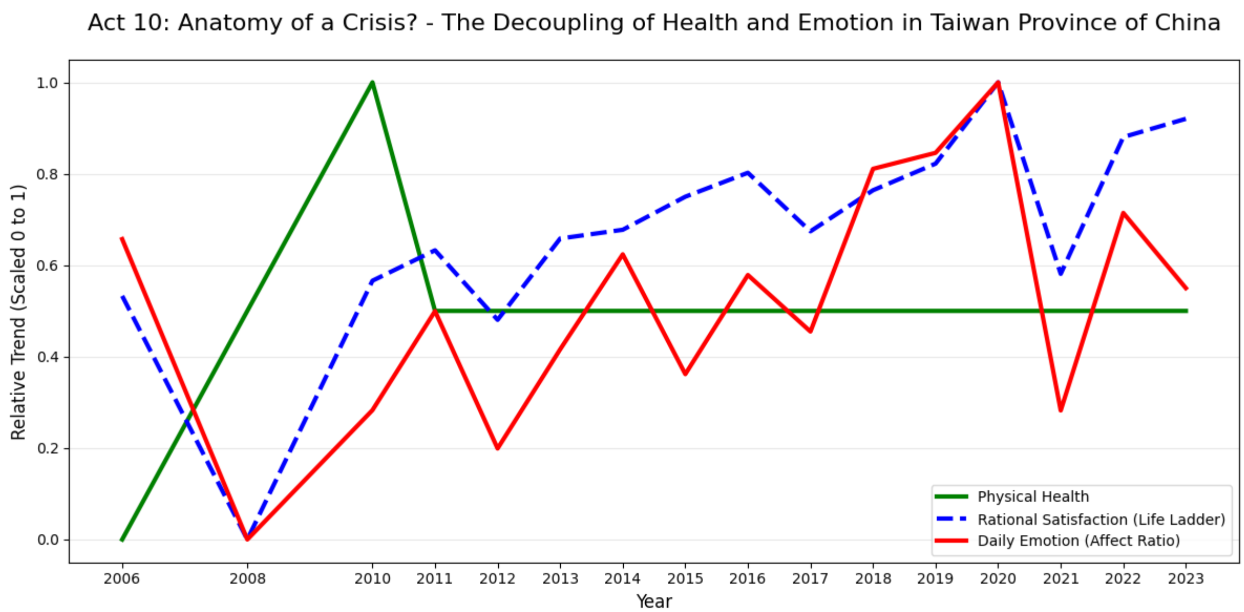
- Healthy Life Expectancy (0.69): A strong link, suggesting that in Kazakhstan, physical health improvements are actually translating into better daily emotional states.
- Log GDP per capita (0.59): Economic wealth has a significant positive impact on daily joy.
- Life Ladder (0.55): Rational life evaluation and daily emotions are well-aligned.

These figures provide statistical confirmation for why Kazakhstan appeared in the 'Utopia' quadrant earlier. Unlike the global 'Silent Crisis' (where health rises but emotions stagnate), Kazakhstan exhibits a synchronized growth between biological longevity and daily happiness. This suggests that the country possesses unique social or cultural buffers that maintain emotional resilience even through the socio-political shocks identified in the longitudinal analysis. All these aspects are worth being further explored with a broader dataset or more granular aspects, such as the age of the respondents to the WHR surveys and the specific answers given.

Finally, for the sake of completeness and transparency, we show the trend for Taiwan and its limited value in terms of longitudinal comparison. Indeed, data for healthy life expectancy is limited to 2006, 2010, and 2011, distorting the measurement scale.

```
# Select Taiwan and prepare the time series
target_country = 'Taiwan Province of China'

df_country = df_clean[df_clean['Country name'] == target_country].copy()
df_country = df_country.sort_values('year')

# Extract unique years for explicit xticks
years_range = sorted(df_country['year'].unique().astype(int))

metrics_to_scale = ['Healthy life expectancy at birth', 'Life Ladder',
'Affect_Ratio']

for col in metrics_to_scale:
    min_val = df_country[col].min()
    max_val = df_country[col].max()
    df_country[f'{col}_Trend'] = (df_country[col] - min_val) / (max_val -
min_val)

plt.figure(figsize=(12, 6))

sns.lineplot(
    data=df_country, x='year', y='Healthy life expectancy at birth_Trend',
    label='Physical Health', color='green', linewidth=3
)
sns.lineplot(
    data=df_country, x='year', y='Life Ladder_Trend',
    label='Rational Satisfaction (Life Ladder)', color='blue', linewidth=3,
linestyle='--'
)
sns.lineplot(
    data=df_country, x='year', y='Affect_Ratio_Trend',
    label='Daily Emotion (Affect Ratio)', color='red', linewidth=3
)

plt.title(f'Act 10: Anatomy of a Crisis? - The Decoupling of Health and Emotion
in {target_country}', fontsize=16, pad=20)
```

```
plt.xlabel('Year', fontsize=12)
plt.ylabel('Relative Trend (Scaled 0 to 1)', fontsize=12)

# Explicitly set ticks for every year in the country range
plt.xticks(years_range)

plt.legend(loc='best')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



Conclusions

Our research challenged the assumption that living longer automatically leads to living better. By engineering the Health-Emotion Gap metric, we quantified a growing global divergence: biological longevity is improving almost everywhere, but daily emotional well-being is not keeping pace. Through correlation analysis, we proved that while wealth and health drive rational life evaluations (Life Ladder), they have almost no significant link to daily joy or stress. This confirms that modern healthcare is a necessary foundation, but not a sufficient guarantee, for happiness.

Our investigation began with a challenge to the Linear Myth, questioning whether the global rise in biological longevity was translating into a proportional increase in daily joy. What we discovered was a significant 'Plot Twist': a statistical decoupling where wealth and health drive rational life evaluations but fail to predict emotional equilibrium. By Mapping the Crisis, we identified a global landscape dominated by the Silent Crisis archetype—nations that have mastered physical survival but remain trapped in emotional stagnation. Through our **Regional Hierarchy**, we pinpointed the Middle East, North Africa, and Western Europe as the primary sites of this disconnect.

By mapping nations into quadrants, we identified a massive cluster of countries trapped in a Silent Crisis: nations like Türkiye and many in Western Europe that have achieved medical success but suffer from severe emotional deficits. Our Anatomy of a Crisis deep-dive into Türkiye provided a longitudinal look at how biological stability can mask profound economic and emotional trauma. However, the journey also led us to the **Quest for Utopia**, where we explored deviant cases like Vietnam and Kazakhstan.

Our deep-dives into specific nations show that biological health is remarkably stable, while emotions are highly volatile, sensitive to inflation, banking crashes, and political shocks. Conversely, deviant cases like Kazakhstan and Vietnam show us that high emotional resilience is possible, albeit the two countries offer very different overviews of their 'Utopias'. For Vietnam, the visualisations show how such measurement reflects a specific point in time, that is a euphoric moment of happiness in the aftermaths of the COVID-19 Pandemic. On the contrary, Kazakhstan is a fully deviant case, for which correlation of the `Affect_Ratio` presents a significanlty higher than average correlation with the other indicators, including `Healthy Life Expectancy`. Kazakhstan provides thus a primary case for further research, aimed at understanding whether its cultural peculiarities are the bases of such an emotional resilience or if its social strategy represents a model for other countries.

Our results suggest that the 'Medical Triumph' of the 21st century is incomplete. To close the gap, future policy must shift from purely biological outcomes to structural buffers against the daily economic and social stressors that modern longevity cannot cure. Indeed, the fundamental revelation of this study is the statistical collapse of the Linear Myth. While global investments in healthcare and economic infrastructure have successfully driven a 'Medical Triumph', extending biological life expectancy across every macro-region, these gains have failed to translate into a proportional increase in daily emotional joy. Our analysis proves a profound 'Plot Twist': wealth and health are excellent predictors of how people rationally evaluate their lives, but they share a negligible correlation of only **0.19** with actual daily emotions. This confirms that a high-functioning healthcare system is a necessary foundation for society, but it is not a sufficient guarantee of human happiness.

Our mapping of global archetypes identifies the Silent Crisis as the most pervasive modern condition, particularly within the Middle East, North Africa, and Western Europe. These regions demonstrate that even with advanced longevity, populations remain highly vulnerable to emotional volatility triggered by macroeconomic and socio-political shocks. As seen in the longitudinal study of Türkiye, biological health remains remarkably stable even as daily emotions crash due to external factors like hyperinflation. This suggests that the 'Longevity Economy' is currently incomplete, as it focuses on the quantity of years lived while leaving the quality of daily emotional experience exposed to systemic instability.

However, the deviant case of Kazakhstan offers a vital counter-narrative and a roadmap for future research. Unlike the global trend of decoupling, Kazakhstan exhibits a synchronized growth between health, wealth, and daily joy, maintaining emotional resilience despite severe socio-political trauma. This 'Resilient Outlier' proves that the Health-Emotion Gap is not an inevitability. Future global policy must therefore shift its focus from purely biological outcomes toward creating structural buffers (social, cultural, and economic) that protect emotional well-being of elderly people, ensuring that the triumph of living longer finally becomes the triumph of living better.

```
df_clean.to_csv('world_happiness_processed.csv', index=False)
```

```python
df_agg.to_csv('world_happiness_aggregated_2021_2023.csv', index=False)
```

```python
df_clean.to_csv('world_happiness_clean.csv', index=False)
df_agg.to_csv('world_happiness_agg.csv', index=False)

from google.colab import files
files.download('world_happiness_clean.csv')
files.download('world_happiness_agg.csv')
```

```
<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>
```